

When Crisis Hits, Does Experience Matter?

Evidence from COVID-19 vaccination

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Abstract

Politicians often emphasize prior tenure in office and the political experience it brings to appeal to voters, who value this attribute. However, the role of political experience in determining policy outcomes remains overlooked by the literature, particularly during times of crises, when it becomes especially salient to voters and consequential to policy. This article examines how time in office affects policy during a crisis using data on COVID-19 vaccinations across Brazilian municipalities in 2021. It shows that vaccination rates were 11% higher in cities governed by incumbent mayors reelected in November 2020—who had extensive experience managing the pandemic—compared to cities that elected newcomers. These effects cannot be attributed to bureaucratic continuity alone or differential reelection incentives. They are also concentrated in areas severely affected by the pandemic in 2020, supporting the argument that early crisis management experience, through learning-by-doing, significantly enhanced mayors' performance during the vaccine rollout.

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“Experience is what makes the difference in a crisis” declared incumbent mayor and reelection candidate Bruno Covas in a debate during the 2020 election in São Paulo, Brazil’s largest city. At first, this might seem like a standard claim for an incumbent. Politicians often invoke their time in office—and the political experience it brings¹—to appeal to voters; and voters value it (Abramowitz, 1988; Portmann, 2022). However, Mr. Covas had never held elected office before, having become a mayor only two years earlier when the incumbent resigned to run for governor.² His claim was solely based on the fact that his brief tenure coincided with the first eight months of the pandemic, making him the only one with the “right” experience in that environment.

The prevailing argument linking political tenure to policy performance is *learning-by-doing*: incumbents acquire unique skills that allow them to perform better in the future (Dal Bó and Rossi, 2011; Miquel and Snyder Jr, 2006), especially compared to newcomers whose experiences outside the office might not transfer (Lewis, 2007).³ Despite its potential welfare consequences, the topic has received surprisingly limited attention in political science.⁴ Recent work instead emphasizes the bureaucracy as the main driver of policy outcomes, focusing on incentives and monitoring (Grossman and Slough, 2022; Gulzar and Pasquale, 2017; Raffler, 2022).

Evidence on the incumbent’s experience itself is mixed: while time in office might improve effectiveness (Krause and O’Connell, 2016; Miquel and Snyder Jr, 2006), it can also lead to clientelism, corruption, or policy drift (Chaudhuri et al., 2024; Coviello and Gagliarducci, 2017). More importantly, existing research does not differentiate between “normal” times and periods of stress in the public sector, such as financial crises, natural disasters, or pandemics, precisely the times when the know-how acquired in office is both more unique and consequential for policy.

¹In this article, we use the term political experience to refer to the set of practical knowledge uniquely acquired during an incumbent’s term in office, which is relevant for performance in subsequent administrations in the same office. This encompasses, but is not limited to, skills such as navigating political and bureaucratic networks, coordinating efforts across departments and other government levels, familiarity with administrative procedures, and the ability to select cabinet members and bureaucrats more effectively.

²Mr. Covas was the vice-mayor on the ticket elected in 2016.

³When studying local elections during the COVID pandemic in France, Morisi et al. (2024) state that “voters might consider that incumbents will perform better simply because they have had several years of experience with the local administration and can swiftly mobilize all the available local resources.”

⁴Perhaps because addressing this question empirically is complicated by how positive selection, political experience, and reelection incentives are intertwined (Alt, Bueno de Mesquita, and Rose, 2011; Ferraz and Finan, 2011).

Accordingly, this article addresses this gap by examining the effects of continuous time in office on policy implementation during a time of crisis. It studies COVID-19 vaccinations across Brazilian municipalities in 2021: the year that marked both the initial vaccine roll-out and the beginning of the four-year term of mayors elected at the end of 2020.

Brazilian mayors have extensive control over public health care services that range from basic home visits to complex hospitalizations. During the pandemic, mayors managed both the local containment measures in the initial waves and the vaccination effort after 2021.⁵ In both these tasks, they faced unprecedented challenges in allocating limited physical and human resources to better protect their constituencies. More importantly, they had to appoint, mobilize, monitor, and motivate bureaucrats at all levels, and across all areas, to successfully execute non-routine tasks, often working as volunteers and in extra shifts.

In this context, we answer the question of whether mayors who managed their cities in the first eight months of the pandemic (2020)—and were reelected—were better prepared than newcomers to run the vaccination effort that started soon after the new mayors took office in 2021.

We start by estimating the effect of continuous time in office using a Regression Discontinuity (RD) design. Here, we compare municipalities where an incumbent narrowly won reelection in 2020 (treatment) to those where they narrowly lost to a newcomer (control). By providing a quasi-random assignment of the winner, the RD reduces potential bias in this comparison, which could come from location-specific factors that jointly influence both vaccination rates and electoral outcomes. Moreover, it rules out positive selection as an explanation, since in close elections the incumbent's win or loss is, by assumption, unrelated to prior performance.

Our results show that reelected incumbents vaccinated 10.6% more than newcomers in similar locations by the end of 2021, which resulted in 7% fewer deaths in 2021–2022 in treated locations. We interpret these positive effects of continuous time in office primarily as the consequence of a gap in political experience between the candidates, especially experience acquired

⁵While the central government was primarily responsible for procurement and allocation of doses across municipalities, the distribution to citizens was carried out by municipal administrations and their bureaucrats. Mayors generally placed a high priority on vaccination and were directly involved in critical stages of the process, including the creation and enforcement of guidelines such as the time between doses and the prioritization of age groups.

during a crisis.

Nevertheless, this estimate still captures the bundled effect of reelecting an incumbent (the only candidate with continuous time in office) and other traits correlated with incumbency. We do three things to assess the relative importance of potential compound treatments. First, we show that in close races, reelected incumbents and newcomers are indistinguishable across a wide range of observed characteristics, suggesting that these traits do not drive our findings.

Second, using a series of tests in Appendix C, we examine whether unobserved candidate competence could bias our estimates by acting as a compensating differential in the close-election design (Marshall, 2024). In short, our evidence suggests that it is unlikely that incumbents and newcomers significantly differ on this dimension in our context.

Third, we address the role of reelection incentives. As reelected incumbents are term-limited in Brazil, only newcomers could run again in 2024. That said, if the observed vaccination gap between incumbents and newcomers is driven by these differing incentives, it is because the latter perceive lower vaccination rates as electorally advantageous. We argue that this would be more likely in places with right-leaning electorates or those governed by right-wing mayors, which would lead to larger effects in these locations. President Bolsonaro’s denialist rhetoric (Araújo and Kerler, 2024; Leal, 2024) contributed to greater vaccine hesitancy among far-right voters,⁶ and led right-wing mayoral candidates to place less emphasis on the pandemic in their campaigns.

Nevertheless, the results do not support this narrative. In fact, the effects are much higher within the group of municipalities governed by left-leaning parties, which suggests a more nuanced interpretation of the results: effective vaccination outcomes seem to reflect not only the mayor’s ability stemming from experience, but her willingness, which is shaped by the party line.⁷ While higher experience leads incumbents to consistently outperform newcomers, this gap is weaker among right-wing politicians, where incumbents and newcomers may be similarly less inclined to promote vaccination due to party cues. For left-wing mayors, who are more incen-

⁶Despite increased polarization, overall vaccine acceptance in Brazil remained among the highest globally, even during the pandemic (Datafolha, 2021; Lazarus et al., 2022).

⁷In this context, future career prospects within the party organization may play a central role in shaping policy choices of term-limited mayors.

tivized to focus on this policy effort, the ability gap between incumbents and less experienced newcomers results in a higher policy impact.

We also provide additional support for our claim that the incumbent’s recently acquired crisis-related experience is the main driver of our effects. We first do so by examining the role of bureaucratic continuity as a potential explanation. Bureaucratic turnover has been widely documented in Brazil whenever there are changes in either the mayor or the party leading local administrations (Colonnelli, Prem, and Teso, 2020; Frey and Santarrosa, 2024; Goyal and Sells, 2024), with downstream implications for public services (Akhtari, Moreira, and Trucco, 2022; Toral, 2024). Here, we ask whether the effects could be uniquely explained by higher bureaucratic continuity under incumbents (and perhaps the accumulation of bureaucratic experience) rather than the mayor’s continuity. Using an alternative sample of cities where mayors did not run for reelection, but their parties did (the *Party Turnover* sample), we use the RD to estimate the effect of an incumbent’s party victory on vaccination, but with a new mayor who lacks continuous office experience, just like a newcomer. We then use this sample to benchmark the impact of bureaucratic turnover (or continuity) on vaccination, which helps us interpret the findings in our main sample (the *Mayor Turnover* sample).

Overall, we find similar patterns of bureaucracy changes in both samples: when either incumbent mayors or incumbent parties lose, dismissals and new hires of health care workers occur at comparable rates. More importantly, vaccination rates increase only in the Mayor Turnover sample, while no significant effects are observed in the Party Turnover sample. This contrast indicates that bureaucratic reshuffling alone is not what drives these higher vaccination outcomes.

In fact, our findings are better interpreted as reflecting the joint continuity of incumbents and bureaucrats, consistent with the anecdotal evidence presented later that a substantial share of the know-how accumulated by mayors during the pandemic was tied to how they selected, deployed, and coordinated their bureaucrats. While precisely disentangling these components lies beyond the scope of this article, we provide suggestive evidence that bureaucratic continuity accounts for approximately 15% of the vaccination gap between incumbents and newcomers.

Finally, we show that the effects come almost entirely from the subset of municipalities with higher COVID incidence in 2020, when mayors faced the worst of the pandemic. Such a result most likely reflects transferable skills that mayors developed on the job during their first term.

Municipal Governance and COVID-19 Vaccination in Brazil

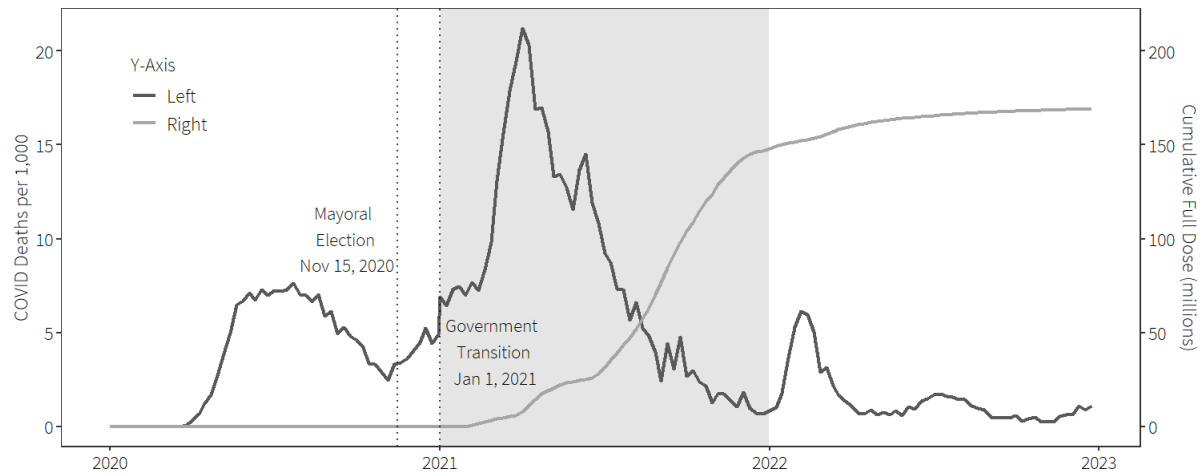
Brazil is a three-tier federative republic with autonomous federal, state, and municipal governments. Municipalities are the lowest level of autonomy, run by democratically elected mayors. In the period covered in the paper, Brazil had 5,570 municipalities. Municipal elections occur every four years in all locations on the same day. Mayors are elected by plurality in all municipalities except 95 cities with more than 200,000 registered voters, which hold a runoff between the top two candidates.⁸ All mayors are limited to two consecutive terms in office.

Municipal governments are responsible for delivering key public services, including education, health care, social assistance, transportation, and sanitation. Most health care is provided through the public universal health system (SUS), although higher-income households often use private insurance. The public health system is decentralized: the federal government concentrates on funding and regulation, while states and municipalities focus on providing care (Brazilian Constitution, Art. 23 and 30, VII). Municipalities offer diverse health services ranging from basic preventive care (e.g., vaccinations and home visits) to complex care (e.g., hospitals and ICUs). Especially during the pandemic, the public sector monopolized the distribution of the COVID-19 vaccine until June 2022 (Law 14,125 of 2021), and municipal health clinics administered nearly all vaccinations throughout 2021.

The 2020 mayoral race took place on November 15, and elected mayors assumed office on January 1, 2021—between the first and second nationwide waves of COVID-19 and only a few weeks before vaccines became available in the country (Figure 1).

⁸Runoff elections are held when no candidate achieves a majority of valid votes (Brazilian Constitution, Art. 29, II). Our main sample conservatively excludes this subset of large municipalities to ensure we only compare cities with similar electoral calendars. Table B.7 (appendix) shows that their inclusion has no meaningful impact on the results.

Figure 1: COVID Deaths and Vaccines in Brazil



The shaded area represents the year 2021, which is the focus of this analysis. Data on COVID-19 infections and vaccines come from [MS \(2020\)](#) and [MS \(2024\)](#), respectively.

The second wave, which occurred during the first half of 2021, resulted in a substantially higher total number of deaths than the first wave, despite having a lower mortality rate compared to the early infections of 2020 (see Appendix Figure [B.2](#) for mortality rates). Figure [B.1](#) (appendix) also presents an index tracking hospital occupancy, and shows that the public health system faced severe strain during February and May 2021. For instance, in March, 24 out of 26 states had ICU occupancy rates exceeding 80%. The expansion of vaccine distribution throughout 2021 coincided with a sharp decline in total deaths, mortality rates, and hospitalizations.

Vaccines in Brazil are a matter of national pride ([Gramacho et al., 2024](#)). The country has one of the largest public vaccination programs in the world, offering free immunizations for more than 20 diseases nationwide. In 2019, 88% of the population believed that vaccines were important for children, one of the highest acceptance rates worldwide ([Figueiredo et al., 2020](#)).

However, vaccination became increasingly politicized after the 2018 election of President Jair Bolsonaro. During the pandemic, he downplayed the severity of COVID-19, questioned vaccines, opposed social distancing measures, and promoted ineffective treatments such as hydroxychloroquine ([Araújo and Kerler, 2024](#)). In response, Congress affirmed the authority of

local governments to adopt health measures to fight the pandemic (Law 13,979, 2020), and the Supreme Court blocked federal interference in local COVID-19 containment policies (Decision ADPF 672, 2020). As a result, mayors took an active role in enforcing measures such as business closures and social distancing ([Chauvin and Tricaud, 2024](#)).

The 2021 immunization effort followed the same decentralized structure, with municipalities responsible for distributing vaccines to households. New mayors that started their 4-year tenure in 2021 faced similar challenges as those faced by the 2017-2020 administrations when implementing containment policies.⁹ Frustrated by delays in federal vaccine procurement, mayors in several state capitals even sought a supply deal with the Butantan Institute ([Exame, 2021](#)), and over 1700 cities tried to form a consortium to purchase vaccines directly ([Brasil 247, 2021](#)). However, the Ministry of Health ruled that any vaccines acquired by local governments or private actors would be centralized and redistributed according to technical criteria ([O Tempo, 2021](#)).

In practice, the allocation of vaccines followed a needs-based approach. The Ministry of Health distributed doses in proportion to the size of priority groups.¹⁰ Given the participative logistic plan and the technical criterion for proportional vaccine assignment, it is unlikely that the distribution of vaccines across municipalities was influenced by political discretion.¹¹ Despite Bolsonaro's anti-vaccine rhetoric and tensions among different government levels, public demand for vaccination remained high. By the end of 2021, 80% of the eligible population (177 million people) had completed the full vaccination schedule ([MS, 2021a](#)).

However, coverage varied substantially across municipalities. By the end of 2021, the standard deviation in vaccination rates was 16 percentage points, and one-quarter of municipalities had coverage below 67%. In this context, these differences largely reflect variation in the quality

⁹There is extensive anecdotal evidence that mayors were keen to start vaccination. For example, the National Mayors Front characterized the federal procrastination in sourcing vaccines as unreasonable. The president of the National Confederation of Municipalities, Gladimir Aroldi, also expressed concerns about possible inequities arising from a lack of coordination among local vaccination strategies ([Exame, 2020](#)).

¹⁰The distribution plan was defined in coordination with the National Council of Health Secretaries (CONASS) and the National Council of Municipal Health Secretariats (Conasems), institutionalized bodies representing all states and municipalities. The Ministry of Health shared technical reports with these councils specifying allocation quotas by municipality and state ([MS, 2021b](#)).

¹¹In the appendix, Table [B.11](#), we confirm this assessment by showing that our political effects on vaccination rates are not driven by differential allocation across municipalities.

of municipal policy implementation.

Mayors and the Immunization Effort. Since the supply of vaccines was beyond mayors' control, most of their influence came in two forms. First, they planned immunization campaigns: setting intervals between doses, prioritizing age groups, communicating eligibility rules and appointment schedules, deploying mobile units to remote areas, and managing storage and logistics to avoid wastage. Second, and more importantly, implementation required a large-scale reallocation of human resources across municipal governments and civil society. The urgency and scale of the rollout led many cities to create task forces (*mutirões*), often involving extra shifts and mobilization of hundreds or even thousands of public employees, as in Salvador, BA ([Salvador, 2021](#)). These efforts also drew on staff beyond the health sector, including traffic management and public security, as illustrated by Corumbá, MS ([Corumbá, 2021](#)).

Mayors also treated vaccination as a priority.¹² In the state capital, Porto Alegre, RS, the mayor's cabinet mobilized different areas of the bureaucracy to volunteer extra hours in the immunization effort ([Porto Alegre, 2021](#)). In Jundiaí, SP, the mayor organized an event to honor hundreds of volunteers in these campaigns, including retired health care workers ([Jundiaí, 2021](#)).

Outperforming municipalities also streamlined vaccinations by setting up temporary vaccination sites, implementing online pre-registration, and extending hours to weekends, as in São Paulo, SP ([São Paulo, 2021](#)). By the end of each day, some cities used expiring doses to proactively vaccinate passersby and avoid waste ([O Globo, 2021](#)). Non-standardized documentation requirements for the vaccine across the country also influenced the effectiveness of the campaigns, since excessive requirements confused citizens and caused long lines, diverting health workers' time that could be proactively seeking unvaccinated citizens ([BBC News Brasil, 2021](#)).

Logistical decisions were consequential. Poorly managed lines in public health clinics, such as those in Belford Roxo, RJ, hampered vaccination intake among elderly citizens ([G1, 2021](#)). Some municipalities also failed to disseminate information about vaccination dates and locations in remote neighborhoods, all actions that required bureaucratic mobilization. This led to unused

¹²For example, see the statement of the new mayor in Valinhos, SP ([Valinhos, 2021](#)).

doses being returned to the state pool, even in locations without full coverage ([G1, 2021](#)).

Political Experience and COVID Vaccinations

This paper investigates whether administrative experience acquired during a time of crisis affects incumbents' subsequent performance in policy provision.

There is a large literature that examines what determines politicians' effectiveness in developing contexts ([Grossman and Slough, 2022](#)). Much of this work focuses on the incentives that lead incumbents to put effort into public good provision. Particularly, the literature focuses on the role of accountability mechanisms, emphasizing how elections allow voters to monitor and reward politicians for their performance ([Ashworth, 2012](#); [Duggan and Martinelli, 2017](#)). Yet because accountability systems often function poorly in developing nations, public good provision is also shaped by practices widely viewed as detrimental, such as clientelism and corruption, both well documented in Brazil ([Frey, 2019](#); [Hidalgo and Nichter, 2015](#); [Nichter, 2018](#)).

Recent work also highlights how politicians' descriptive traits shape representation in terms of substantive policy outcomes. Particularly in the Brazilian context, policies are influenced by politicians' characteristics such as gender ([Chauvin and Tricaud, 2024](#)), race ([Bueno and Dunning, 2017](#)), education ([Desai and Frey, 2023](#)), or professional background ([Novaes, 2024](#)).

This article shifts the focus away from **politician's incentives** to their **ability** to implement public services. While some work has examined the role of incumbents' characteristics, such as education ([Carnes and Lupu, 2016](#)), on policy outcomes, much more emphasis has been placed on how the incumbent's ability to "get things done" is shaped by their relationship with the bureaucracy ([Grossman and Slough, 2022](#)). The performance of bureaucrats often determines policy outcomes ([Slough, 2021](#)), and in doing so mediates the politician-voter relationship, directly influencing incumbents' electoral prospects ([Dasgupta and Kapur, 2020](#); [Martin and Raffler, 2021](#); [Slough, 2024a](#)). Accordingly, public service delivery is typically more effective when institutions enable politicians to monitor bureaucratic performance and credibly claim credit for outcomes

([Gulzar and Pasquale, 2017](#); [Raffler, 2022](#); [Slough, 2024b](#)); and when the agents' policy priorities and goals are aligned with the politician's ([Toral, 2023](#)).

By answering our initial question, we highlight an underexplored dimension within this literature: how political experience shapes politicians' ability to deliver policy, particularly expertise acquired during crises. Whereas bureaucratic expertise has long been recognized as central to implementation, less attention has been paid to the policy-relevant skills that politicians themselves accumulate over time. Although the performance of incumbents is likely to depend on a variety of qualities, some critical skills are learned only on the job ([Krause and O'Connell, 2016](#); [Miquel and Snyder Jr, 2006](#)), and expertise acquired in other careers might not always transfer to elected positions in public administration ([Lewis, 2007](#)).

Despite its intuitive appeal, the idea that political experience improves policy is not self-evident. The literature provides limited evidence that experience drives good policies, even in normal times. While some studies find that tenure increases effectiveness ([Krause and O'Connell, 2016](#); [Miquel and Snyder Jr, 2006](#)), others show that experience can facilitate clientelism, corruption, or policy drift ([Chaudhuri et al., 2024](#); [Coviello and Gagliarducci, 2017](#); [Klašnja, 2015](#)). This might be due to the difficulty in empirically disentangling the effects of positive selection and experience on performance ([Alt, Bueno de Mesquita, and Rose, 2011](#)). Moreover, experience can be a "double-edged sword" ([Tsur, 2022](#)), and learning by doing could also lead to malfeasance.

These considerations bring us back to our central question, for which the Brazilian case provides an empirical test: did the experience of managing the initial eight months of COVID-19 improve mayors' performance in the 2021 vaccination campaign? We assess the question in a crisis environment where a relevant dimension of on-the-job learning is the dynamic relationship between incumbents and the bureaucracy. The Brazilian context described above provides several examples of cities where the success of vaccination efforts resulted from the mayor's ability to mobilize, monitor, and effectively motivate personnel to ensure that vaccines were distributed broadly and quickly. This required high-level decisions such as the appointment of leadership positions and the creation of immunization plans, as well as the coordination of the logistical

effort of street-level bureaucrats in the delivery of vaccines.

We also argue that, due to both the urgency and the scale of the policy intervention here, the type of experience that might boost the performance of politicians is likely to be unique. If expertise is acquired through repetition, then mayors in this position between March and December of 2020 should be uniquely prepared for the 2021 vaccination effort. During the first months of this crisis, they faced similar challenges in managing resources during the initial waves of infections, particularly in allocating limited physical resources within their jurisdictions, such as ICU beds or ventilators; and working to retain, motivate, and mobilize health care workers to safely perform non-routine tasks under pressure.¹³

In the next section, we present both our data and empirical design, describing how we use them to isolate the impact of crisis experience on vaccine coverage from alternative explanations. We then present and discuss our findings in the following section.

Data and Empirical Design

This article uses vaccination data from Brazilian municipalities to examine the effect of continuous time in office on public good provision in the context of crisis response. The empirical strategy benefits from the timing of the local electoral cycle, given that the nationwide mayoral elections happened on November 15, 2020, precisely between the first and second waves of COVID-19 infection (Figure 1). All mayors elected in November took office on January 1, 2021, just 17 days before the country administered its first COVID-19 vaccine shot.

This context creates the opportunity for a comparison in vaccination rates between municipalities where an incumbent mayor was reelected in 2020 and those that elected a challenger. While the latter group had a newcomer in charge of the vaccination, the former had a mayor with recent experience in office, and, more importantly, experience acquired during eight months of intense COVID-19 infections when mayors had control over several mitigation policies.

¹³This was due not only to COVID infections, but also the backlog of other health conditions that went untreated during that period.

Nevertheless, there are a few reasons why this comparison might not isolate the impact of time in office on vaccinations. First, municipalities that reelected an incumbent might fundamentally differ from those that elected a challenger, as local characteristics might affect both the reelection rate and the quality of local public services. Second, reelected incumbents are different from challengers not only on their recent office experience but also on their “proven ability”, i.e., the fact that they have won an election after revealing their performance in office (Frey, 2021).

We address these two challenges using a regression discontinuity design (RDD), where we compare municipalities where an incumbent mayor was reelected in 2020 in a close race to those where the incumbent lost. The RD mitigates concerns regarding the influence of municipal characteristics that might confound the results and potential bias coming from proven ability. Since the incumbent’s victory (or defeat) in this design comes from the quasi-random assignment of the election winner, it cannot be attributed to the incumbent’s recent performance in office.¹⁴ Accordingly, we estimate the following equation:

$$Vacc_i = \beta_0 + \beta_1 Inc_i + \beta_2 D_i + \beta_3 Inc_i D_i + \epsilon_i \quad (1)$$

where for municipality i , the binary variable Inc_i indicates whether the incumbent won the last election, and D_i is the difference between the election winner and the runner-up in vote percentage.¹⁵ The treatment effect on the vaccination rate $Vacc_i$ is given by β_1 . As usual, regressions are weighted by the triangular kernel, and the sample is restricted to values of D_i that fall within an optimal bandwidth (Calonico et al., 2019). In the appendix, we show that both the density of the observations (Figure B.3) and municipal covariates measured pre-treatment (Table B.1 and Table B.2) are continuous at the cutoff (i.e., commonly referred to as McCrary and Balance tests). All election data was obtained from the Brazilian Superior Electoral Court (TSE, 2016-2020).

¹⁴The RD effect is only identified for a sample of municipalities where the incumbent mayor actually ran for reelection and finished in the top 2. Please see the article’s conclusion for a discussion on the external validity of these findings.

¹⁵As it is usual in the literature on close-election RD, we restrict our sample to races where an incumbent was one of the two leading candidates.

Although our main results are presented without covariates,¹⁶ in the appendix we show that they are robust to the inclusion of a range of pre-determined municipal and mayoral characteristics, including population and density, GDP per capita, COVID-19 incidence in 2020, municipal health personnel, campaign spending limits, mortality rate, 2018 vote share for Bolsonaro, and mayoral attributes (race, education, gender, occupation, and party). A full description of these variables is provided in Table B.2.

The main outcome variable is the vaccination rate in 2021 in each municipality.¹⁷ This data was obtained from the Health Ministry (MS, 2024), and the variable is defined as the percentage of the local vaccine-eligible population¹⁸ that has received the vaccine during that period. We focus on the full dose of vaccination in 2021, but also show the results for the first dose, and for later periods.¹⁹

Accordingly, β_1 identifies the treatment effect of having a reelected incumbent as mayor vs. a newcomer. In close election designs that use politician characteristics to assign treatment status, estimates should be interpreted as the compound effect of incumbency status and other individ-

¹⁶Covariates do not play a role in treatment effect identification in RDD designs. They are often included to reduce the variance in the estimation.

¹⁷Our dependent variable is constructed from individual-level vaccination microdata that records each administered shot, and includes individual recipients' data such as age, sex, date of application, dose type, manufacturer, and batch. These records are entered by health-unit staff at the point of vaccination and uploaded to a centralized national system, rather than reported in aggregate by municipal executives. Within this system, large-scale manipulation of vaccination numbers by mayors is unlikely. It would require coordinated falsification across health facilities and personnel over time, making it operationally costly. Also, once uploaded, individual vaccination records became visible to citizens through the Ministry of Health's app within approximately ten days, and individuals can request corrections if discrepancies are identified between their physical vaccination card and the digital registry. This creates an additional layer of decentralized monitoring that further constrains systematic misreporting.

¹⁸As a proxy for the target population for vaccination in 2021, in each municipality, we use the number of registered voters in November 2020. This proxy is preferable for two reasons: (i) it provides a more updated measure of the local population than the most recent census data, which is from 2010; and (ii) it closely reflects the vaccine-eligible public, since voting in Brazil begins at age 16 and is mandatory between ages 18 and 70 (and voters older than 70 automatically remain in the registry for the rest of their lives). Vaccines were not approved for ages 12–15 until June 2021 and for ages 5–11 until December 2021. We show in Table B.5 that results are robust to using the 2010 Census total population—obtained from IBGE (2010)—and the number of actual voters present in the 2020 municipal election. Compared to using registered voters, the latter measure might capture the most recent migration movements, but miss potential differences in turnout.

¹⁹The raw vaccination data has observations at the individual-shot level, with additional information on the date and brand of the vaccine taken. We aggregate first, second, and single doses by vaccination site and time period (year or quarter, in different specifications). At the time, the Janssen vaccine was the only single-dose vaccine. Following official statistics guidelines (MS, 2024), we exclude any vaccination record dated before January 17, 2021, the day Brazil administered its first official COVID-19 vaccine shot. This date discrepancy affects fewer than 0.4% of all recorded doses, presumably due to data-entry errors.

ual characteristics that may differentiate incumbents and newcomers at the discontinuity (i.e., when the margin of victory equals zero). Among them, this article aims to emphasize the role of continuous experience on the job, which is higher for reelected incumbents by construction. We show in the appendix (Table B.1 and Table B.2) that incumbents and newcomers do not differ, at the discontinuity, in several other individual characteristics that could potentially impact vaccination outcomes, such as race, gender, education, professional background, and ideology.

Unobserved Competence as a Compensating Differential. We also address a related concern in the design with the role of unobserved competence²⁰ on the effects (Marshall, 2024). It is reasonable to expect that the politician’s unmeasured competence could positively impact vaccinations while being also associated with higher reelection probability.

In the context of the present study, the main concern is whether incumbents that appear in close elections are, on average, more competent than the challengers that they face. This possibility arises only if (i) voters observe and reward candidate competence; and (ii) if there exists a candidate-level incumbency disadvantage that pushes the vote shares of high-ability incumbents down, placing them into close races. In this case, our estimates of the effect of an incumbent’s victory capture ability rather than experience.

We discuss and address this concern in detail in Appendix C, providing three pieces of evidence against this mechanism. First, we regress the incumbency status of 2020 candidates on their vote shares to show that incumbency and votes are in fact positively correlated (Appendix Table C.2).²¹ While this might suggest that incumbents have, in fact, a personal advantage in Brazil, this coefficient might still reflect the combination of a disadvantage plus a (larger) competence boost. Thus, we run a sensitivity analysis (Oster, 2019) to determine under what conditions we would observe a candidate-level incumbency disadvantage—or a negative coefficient in this regression—if we could “control” for competence. Overall, we find that such a candidate-level disadvantage is

²⁰Competence may not be fully captured by measured traits such as education and professional background. We note that the problem of compensating differential in RDs arises precisely with the concern that some characteristic of the candidate, such as competence, is observed and rewarded by voters, but unobserved by the researcher.

²¹We emphasize that the incumbency disadvantage documented by Klačnja and Titunik (2017) in Brazilian mayoral elections is at the PARTY level, and driven by lame duck mayors that cannot run for reelection themselves, instead of the reelection-eligible incumbents that we study here.

extremely unlikely. Incumbency has a strong, positive, and stable relationship with vote shares, and unmeasured confounders such as competence would have to account for more than 100% of the remaining unexplained variation in vote shares for the coefficient to approach zero.

Second, we examine the effect heterogeneity based on the 2016 margin of victory in each locality. If vote shares reflect competence, then mayors first elected in 2016 in a close race are less likely to possess a substantial competence “premium.” In that case, if our vaccination effects were driven by a quality gap between incumbents and challengers—rather than by experience—they should be weaker for this group, which is not the case (Appendix Table C.1).

Third, we show that the decision of eligible incumbents to run in 2020 is uncorrelated with their margin of victory in 2016, which reduces the concern that only the more competent incumbents choose to seek reelection in 2020.²²

Reelection Incentives as a Potential Compound Treatment. Brazil has a limit of two consecutive terms for mayors. This means that reelected incumbents in 2020 could not run again in 2024, when newcomers could. These reelection incentives have been shown to influence other dimensions of public policy implementation, such as corruption (Ferraz and Finan, 2011), and could be driving the differences in immunization rates. The particular concern here is that higher vaccination coverage under reelected incumbents could be driven by the actions of newcomers in contexts where fewer vaccines might improve their likelihood of reelection, as they become more responsive to either the preferences of local voters or the guidance of party leadership.

We assess this alternative narrative by leveraging the fact that the national political environment during the pandemic was polarized around the issue, with right-wing President Jair Bolsonaro championing anti-vaccination arguments on one side and the Left-wing opposition led by the Workers’ Party (PT) on the other. We use the proximity to Bolsonaro as a proxy of anti-vaccine stance for both the average electorate and the mayor’s party in each municipality. We then examine the heterogeneity patterns in the RD effects for locations where newcomers would have more incentives to shirk the vaccination effort, according to this narrative.

²²Overall, 75% of reelection-eligible incumbents run again in 2020.

Bureaucratic Continuity as an Alternative Explanation. This article emphasizes the role of mayoral continuity—and the experience it brings—on vaccination. However, one might be concerned that the higher vaccination rates under incumbents may reflect the continuity in the bureaucracy instead. Recent literature shows that changes in either the mayor or the ruling party in Brazilian municipalities are often accompanied by substantial bureaucratic turnover (Colonnelli, Prem, and Teso, 2020; Frey and Santarrosa, 2024; Goyal and Sells, 2024), which has further consequences for the quality of public services (Akhtari, Moreira, and Trucco, 2022; Toral, 2024).

We address this concern by focusing on an alternative sample of cities where incumbents did not run for reelection in 2020, but their parties did with a new candidate (this is the *Party Turnover* sample).²³ With the same RD design, we compare places where the incumbent party won in a close race to municipalities where it lost and identify the effect of electing a new **party** to office. Here, both treatment and control groups have new mayors with a comparable lack of recent office experience and reelection incentives.²⁴ Thus, it is more likely that the effects observed in this sample are due to turnover in the bureaucracy, which we measure with individual data on the universe of health care workers in Brazil provided by the National Registry of Health Establishments (CNES, 2020-2021). This will inform us about the impact of this dimension on the COVID-19 vaccination effort, allowing us to separate it from the effects of mayoral continuity in our main sample (the *Mayor Turnover* sample).

The validity of this exercise depends on how similar the Mayor Turnover and Party Turnover samples are. Table B.13 (Appendix) shows the similarity between samples in several characteristics of their municipalities, including the bureaucracy profile and their elected mayors. In locations where the 2020 election was close, which is the group used in our estimation, we find no statistically significant difference between the samples in any of the variables.

Finally, we acknowledge that RD designs prioritize internal over external validity due to the local nature of treatment effects. One notable concern is whether one could expect the same

²³In most of these municipalities (69%), incumbent mayors could not run because they were term-limited, not because they decided not to do so.

²⁴Table B.1 (appendix) also shows that several covariates measuring pre-treatment characteristics of municipalities, candidates, and parties are continuous at the cutoff for this sample.

effects of experience in places with less competitive elections. Although the nature of the design limits our ability to advance such claims, we show in the Appendix Table B.12 that the group of cities with close elections in 2020 is similar in several pre-existing characteristics to the rest of the sample. Close-election places seem to be slightly wealthier than the rest, but are similar in size, political profile, previous COVID-19 cases, size of the bureaucracy, and profile of the mayor. These results at least suggest that we should not expect idiosyncratic municipal characteristics to be a strong driver of effect heterogeneity in places with less electoral competition.

Results and Discussion

Table 1 shows the estimated effect of reelecting the incumbent mayor on the 2021 vaccination rates (β_1). Figure 2 has the graphical representation of the results, and the appendix shows their robustness to changes in bandwidths (Figure B.5); the inclusion of covariates and changes in the polynomial (Table B.3); and different outcome variables (Table B.5).

Table 1: Effect of an Incumbent’s Victory on COVID-19 Vaccinations in 2021

	Full Dose	1st Dose	Full in 2021–2022
Treatment Effect	8.883* (4.027)	8.477* (3.848)	9.156* (4.102)
C.I. (95%)	[1.54,17.33]	[1.41,16.49]	[1.65,17.73]
Baseline	83.704	96.501	96.706
Effect over Baseline	10.61%	8.78%	9.47%
Bandwidth	2.74	2.80	2.84
Municipalities	362	367	371

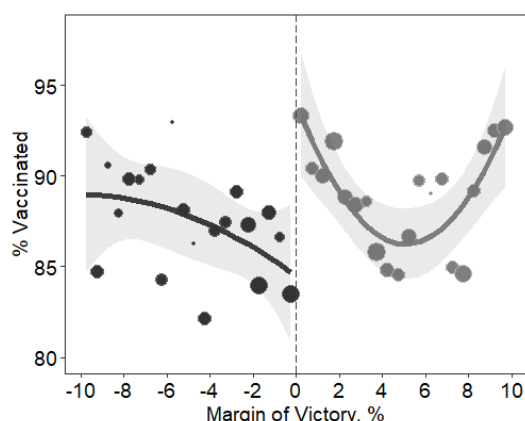
+p<0.1, *p<0.05. The effect corresponds to β_1 in equation 1, for the **MAYOR Turnover Sample**. Standard errors and 95% confidence intervals are bias-robust. The Baseline is the average outcome for the municipalities that elected newcomers (control group) at the cutoff.

Overall, on average, incumbent mayors achieved vaccination rates nearly 9 percentage points (p.p.) higher than newcomers in 2021, a 10.6% treatment effect in terms of full dose coverage. These results are similar for the first dose and full doses, considering the aggregate vaccinations in

2021 and 2022 (column 3). Appendix Table B.6 also shows that municipalities with a reelected incumbent had lower increases in their mortality when compared to those where a newcomer was elected. While the total number of 2021–2022 deaths increased 28% vs. 2019 for the control group, cities with a reelected incumbent saw a lower increase of 18% in the same period.

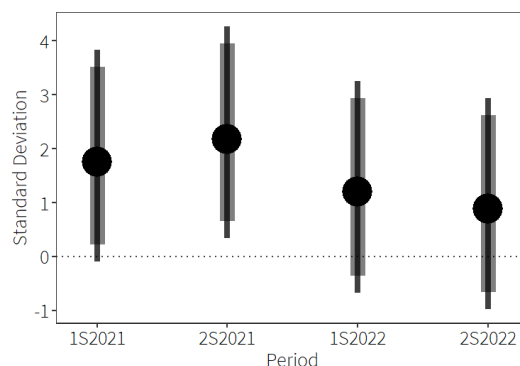
Figure 3 shows that this effect is highly concentrated in the second half of 2021, which is the period of the most substantial ramp-up in the vaccination effort (Figure 1). While in 2022, reelected incumbents continue to outperform their newcomer counterparts, the effects become gradually weaker and lose statistical significance.

Figure 2: RD Representation



Right side: vaccination rates for the victory of an incumbent. Left side: vaccination rates where the incumbent lost. The lines show a quadratic fit.

Figure 3: Effect by Semester



The coefficients correspond to β_1 in equation 1. The thin (wide) bars correspond to the 95% (90%) confidence intervals.

In the appendix, we also show that the effects are driven by the vaccination of municipal residents rather than outsiders. Given that people could cross municipal borders to receive the vaccine, one might be concerned that citizens in control locations (i.e., governed by new mayors) systematically sought vaccination in treated cities due to better management of the vaccination process, thereby inflating the estimated effects. Despite the fact that 20% of the individuals received the vaccine outside of their municipality of residence, our main effect is nearly entirely coming from residents (Table B.4). Similarly, the patterns of border crossing are not correlated with the treatment status of the municipality, as shown by Figure B.4 (appendix).

Finally, Table B.11 (appendix) shows that the effects are not explained by the federal government’s differential allocation of vaccines across cities. Using data on the spatial allocation of doses from Jan. 2021 to Dec. 2022,²⁵ we show that locations governed by reelected incumbents did not receive larger vaccine deliveries compared to those governed by newcomers. On the contrary, the estimated effect on allocation is negative, despite being weak and statistically insignificant.

Experience or Reelection Incentives?

Table 1 shows that vaccination rates are higher under incumbents who have recent experience in office during a period of crisis. However, as discussed on page 15, only newcomers face reelection incentives at the end of their 2024 term. If the prospect of reelection actually reduces effort in vaccination campaigns, this could provide an alternative explanation for our findings.

We assess this explanation by examining effect heterogeneity based on the voters’ ideological profile across cities, as well as the elected mayor’s party ideology, since reelection incentives might lead mayors to become more responsive to the preferences of both voters and party leadership.²⁶

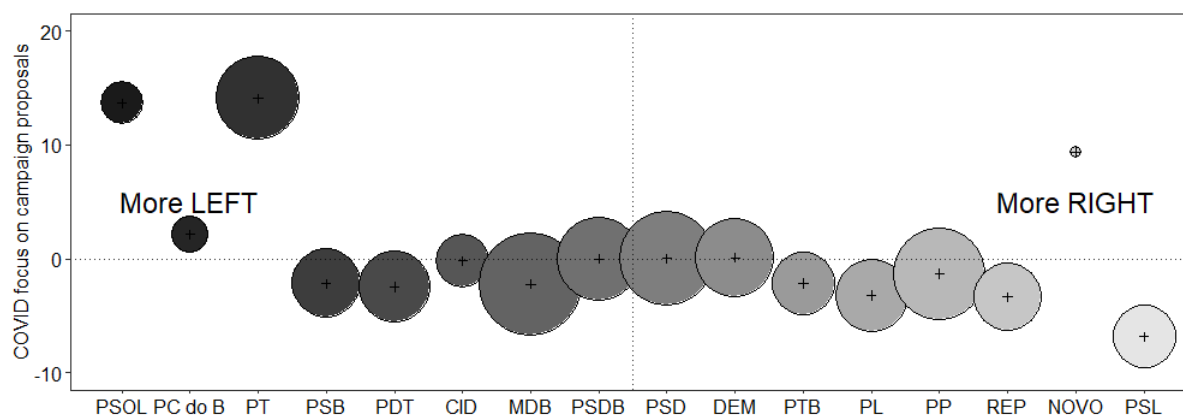
Vaccine and Voter Preferences. From the electorate’s perspective, reelection-eligible mayors might vaccinate less if they think local voters are anti-vaccine. Brazil has had a strong track record in immunization since the 1970s (Gramacho et al., 2024) and maintained one of the highest vaccine acceptance rates globally during the COVID-19 crisis. Lazarus et al. (2022) reports a 90% acceptance rate in June 2021—second only to China among 23 countries—matching the 89% found in a March 2021 survey by Datafolha (2021). Nevertheless, the recent political polarization was associated with vaccine hesitancy in a way that past immunization campaigns were not. President Jair Bolsonaro (2019-2022) actively advanced a denialist discourse that influenced his supporters’ willingness to both avoid the vaccine (Gramacho et al., 2024) and oppose containment measures (Araújo and Kerler, 2024). Accordingly, Datafolha (2021) shows that vaccine acceptance was lower—at 82%—among those who “trust in the words of the President.”

²⁵Unfortunately, MS (2023) only provides the aggregated data on vaccine allocation for the entire period. Thus, we cannot use these data to build separate outcome variables for 2021 or 2022.

²⁶The latter incentive follows the fact that electoral campaigns in Brazil are primarily financed with public funds that are controlled and allocated by national party leaders.

Vaccine and Party Positions. From a top-down perspective, reelection-eligible mayors might vaccinate less if their party is more aligned with the denialist position of the federal executive. Not surprisingly, after examining official documents containing the 2020 mayoral campaign proposals, we find that candidates from left-wing parties were relatively more likely to mention COVID-related issues during their mayoral campaigns in 2020. These proposals are published by all candidates around a month before the election, as required by the Brazilian federal electoral court (TSE), and typically have extended discussions on the candidates' ideas for different areas of the administration. Figure 4 shows the analysis. The x-axis shows the 16 largest Brazilian parties ordered from left to right on an ideological scale.²⁷ The y-axis shows the share of proposals from each party that mention the pandemic,²⁸ as a deviation from the sample average of 22%.

Figure 4: Share of Campaign Proposals that Mention COVID



It uses 12,168 unique candidate proposals across 5,333 Brazilian municipalities. Parties are ordered from Left to Right on the ideological scale found in [Zucco and Power \(2023\)](#). The y-axis shows the % share of campaign proposals that mention COVID, for each party, measured as a deviation from the sample average of 22%. The size of the dots reflects the number of proposals by each party across the country.

With this in mind, if reelection incentives are at least partially driving our results, we should expect larger effects in cities that (a) **have more Bolsonaro supporters**; and (b) **elected Right-**

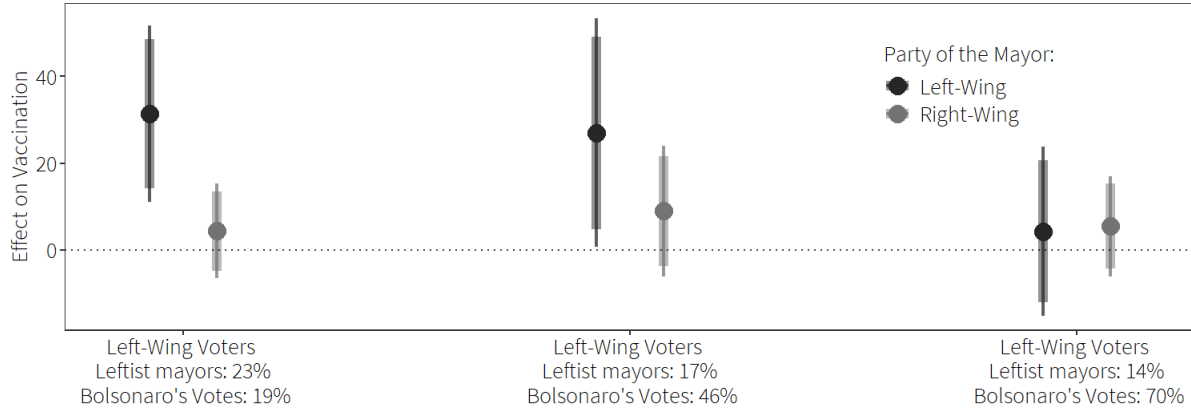
²⁷The sample is based on the 16 parties for which [Zucco and Power \(2023\)](#) provide updated ideological scores. They estimate mean party ideology based on surveys in which each legislator places their own party, as well as others, on a one-dimensional ideological scale. The authors exclude small parties from the questionnaire.

²⁸We consider any document that mentions the words COVID (or Coronavirus) or pandemic.

wing mayors. In both cases, new mayors would put less effort into vaccination relative to incumbents, as they follow either the preferences of the local voters or their party positions.

Nevertheless, Figure 5 shows that this is not the case. The plot shows vaccination effects for six sub-samples: we first divide municipalities into three equally sized groups based on the ideological profile of the median voter: Left, Moderate, and Right.²⁹ Within each group, we then further split municipalities by the partisanship of the elected mayor (Left or Right). Left-wing mayors are a minority in all groups, accounting for 23% in municipalities with a Left-leaning median voter; 17% in the moderate group, and 14% in locations with more Right-wing voters.

Figure 5: Incumbency Effects by the Ideological Profile of Voters and the Mayor's Party



The coefficients correspond to β_1 in equation 1 for each sub-sample. The thin (wide) bars correspond to the 95% (90%) confidence intervals. Estimates are shown in Appendix Table B.8. Here we only use municipalities that elected a mayor from one of the 16 large parties for which we have ideological scores (91% of the sample).

The estimates show a few distinctive patterns. First, all effects are positive and above 4 pp, which shows that experienced incumbents consistently outperform newcomers. Second, not only is there no evidence of larger effects for Right-wing electorates or mayors, but the patterns go in the opposite direction of that predicted by reelection incentives. Third, the municipalities where we observe stronger vaccination differences between incumbents and newcomers are the ones governed by Left-wing mayors, and with Moderate to Left-leaning voters.

Incumbents and the Willingness to Vaccinate. The patterns above also suggest a more

²⁹This is based on the municipal vote share for Bolsonaro in the 2018 Presidential election.

nuanced interpretation of the mechanisms underlying our results. While we have emphasized continuous political experience as a proxy for an incumbent's ability to implement policies, our baseline analysis does not account for potential variation in their willingness to do so. In our context of term-limited mayors, this willingness is likely shaped by the incentives they face from party leadership and future career prospects within these organizations.

As a result, effective policy implementation in this context appears to depend on how the mayor's willingness mediates the impact of their ability. While higher ability leads incumbents to outperform newcomers in all groups, this gap is weaker among right-wing politicians, where incumbents and newcomers may be similarly less inclined to promote vaccination due to party incentives.³⁰ For left-wing mayors, who are more incentivized to focus on this policy effort, the ability gap between incumbents and less experienced newcomers has a higher policy impact.

Mayoral or Bureaucratic Continuity?

On page 16, we discuss how changes in either the mayor or the party in power in Brazilian municipalities could have downstream consequences for both public employment and the quality of policy implementation. Here, we evaluate whether our vaccination results can be uniquely explained by higher bureaucratic continuity under reelected incumbents.

We start by showing the absence of any vaccination effects in the Party Turnover sample (Table 2), a sharp contrast to our main sample results in Table 1. This is consistent with the fact that both candidates in the Party Turnover sample lack recent continuous experience on the job.

We then estimate the effects of the incumbent's victory on several measures of bureaucratic turnover for both samples to check whether differential levels of bureaucratic continuity explain the effect gap between samples. We measure changes in bureaucracy using data on the universe of health care workers in Brazil (CNES, 2020-2021), described in Table 3.

³⁰The fact that we continue to observe positive effects even among less willing right-wing mayors is consistent with anecdotal evidence that many right-wing governors and mayors held more favorable views toward vaccination than the Bolsonaro administration.

Table 2: Effects for the Party Turnover Sample

	Full Dose	1st Dose	Full in 2021+2022
Treatment Effect	-0.562 (5.406)	-2.335 (4.706)	-1.212 (5.767)
C.I. (95%)	[-10.52,10.67]	[-11.18,7.27]	[-11.83,10.78]
Baseline	87.965	99.723	101.090
Effect over Baseline	-0.64%	-2.34%	-1.20%
Bandwidth	4.56	6.03	4.58
Municipalities	160	200	162

+p<0.1, *p<0.05. The effect corresponds to β_1 in equation 1, for the **PARTY Turnover Sample**. Standard errors and 95% confidence intervals are bias-robust. The Baseline is the average outcome for the municipalities that re-elected new parties (control group) at the cutoff.

Table 3: Composition of the Brazilian Health Care Workforce (Municipal Averages), Oct. 2020

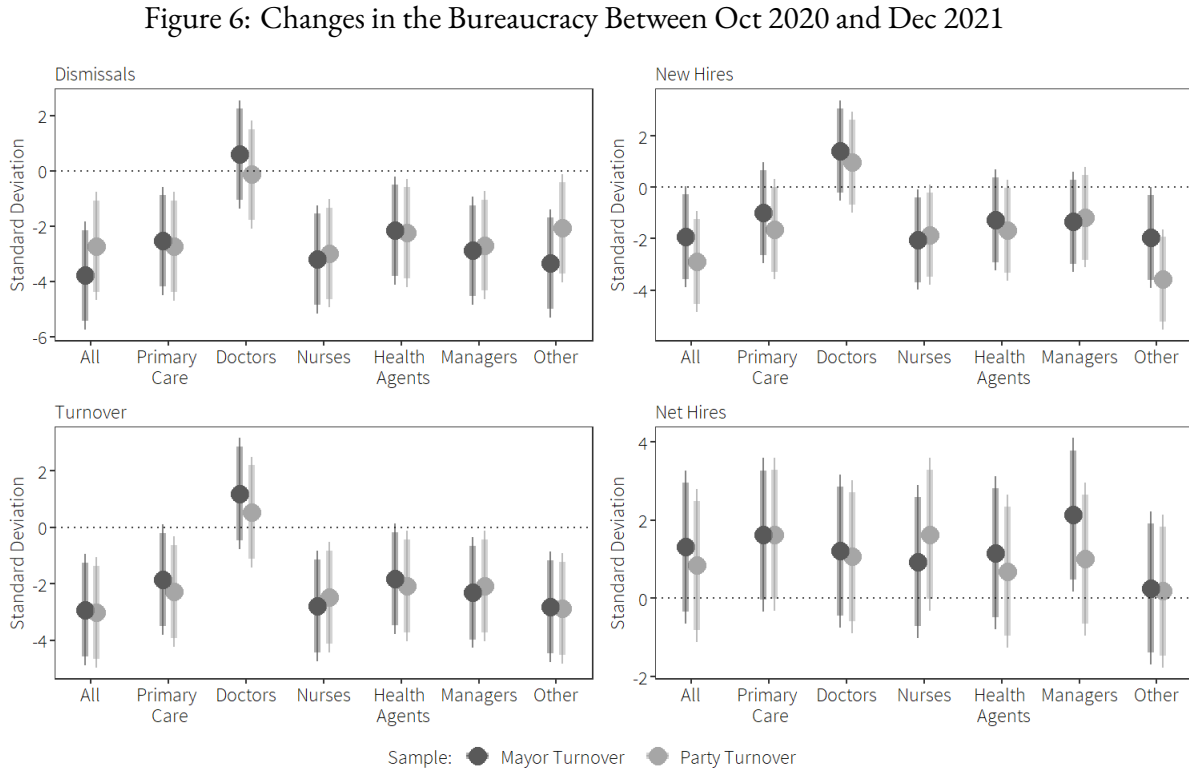
Employer:	PUBLIC SECTOR		PRIVATE SECTOR
	Municipal	State	Private
Total Workers (million)	2.10	0.68	0.70
Municipalities with Presence	5558	2828	3740
% share of workers in each functional category below, by type of employer			
Primary Care	62.02	58.55	28.30
Doctors	8.04	13.42	21.19
Nurses	25.03	40.06	6.91
Health Agents	28.94	5.07	0.20
Managers	2.62	2.81	5.93
Other	35.36	38.64	65.76

The entries show the average % share of workers in each category, computed across municipalities with at least one worker in the respective level or sector of administration. For each sector, the share of workers in Primary Care (Doctors + Nurses + Health Agents), Managers, and Other sum to 100.

The table shows the share of workers in each functional and administrative category in Oct 2020, the month that preceded the municipal election. Out of a total of 3.5 million workers, nearly 80% work under the public universal health system (SUS), and 60% of all do so as employees

of city governments. Of the group of municipal employees, 62% are directly engaged in primary medical care as doctors, nurses, or community health agents.

We then estimate the RD effect of reelecting the incumbent on dismissals and hires of municipal health care bureaucrats between Oct 2020 and Dec 2021.³¹ We also do so for a measure of total turnover (hires plus dismissals) and the net hires in the period (hires minus dismissals). Figure 6 shows the effects for both samples for several job categories.



The coefficients correspond to β_1 in equation 1 for each sample and outcome. The thin (wide) bars correspond to the 95% (90%) confidence intervals.

The plots first show that changes in incumbency, either at the party or candidate level, trigger severe bureaucratic turnover across nearly all relevant job categories in the public health system (the exception is doctors). However, more importantly, the turnover gap between reelected incumbents and newcomers is strikingly similar in both samples. In other words, when the incum-

³¹We classify resignations as dismissals because employees who anticipate being fired during mayoral transitions often choose to resign. In the case of public-sector health workers in Brazil, the effects of changes in mayoral administrations on resignations and dismissals tend to be similar (Toral, 2024).

bent party wins with a new mayor (Party Turnover Sample), the level of bureaucratic continuity is very similar to the one observed for the victory of an experienced, incumbent mayor, even though their performance in vaccination is very different. These patterns help us rule out the narrative that the better vaccination results under reelected incumbents resulted exclusively from the expertise accumulated by the bureaucracy during the early months of the COVID-19 pandemic in 2020, rather than from the incumbent’s own learning during that period.

Does Bureaucratic Continuity Mediate the Effect of Mayoral Continuity? Our institutional setting does not provide a credible counterfactual in which mayoral continuity occurs in the absence of bureaucratic continuity. Thus, our results are better interpreted as potentially reflecting the joint continuity of mayors and health bureaucrats. This interpretation is consistent with the extensive anecdotal evidence presented in the section “COVID Vaccination in Brazil,” which suggests that a substantial share of the experience accumulated by incumbents during the pandemic was closely tied to how they selected, deployed, and coordinated their bureaucrats.

While precisely disentangling these components lies beyond the scope of our empirical strategy, we provide suggestive evidence in the Appendix with two empirical exercises. First, Table [B.14](#) reports the results of a mediation analysis indicating that bureaucratic continuity accounts for approximately 15% of the vaccination gap between reelected incumbents and challengers.³²

Second, Table [A.1](#) shows that the effects are weaker in places with stronger health care bureaucratic capacity (pre-treatment). This suggests that the mayor-bureaucrat relationship matters more for policy in areas where public services are already under stress, and the marginal gains from coordination are likely higher.

Incumbents’ Experience and Crisis Intensity

The results in Table [4](#) provide additional evidence for the unique role of political experience acquired during a crisis. They show how the effect on vaccination varies by the intensity of disease spread in 2020, before the elected mayors took power for the 2021–2024 term. We split

³²We acknowledge the important limitations of this approach, as the key identifying assumption for mediation analysis—sequential ignorability (Imai et al., 2011)—is unlikely to hold in our context.

municipalities into four equally sized groups by their COVID-19 incidence in 2020, estimating incumbency effects on vaccination for each group. The results clearly show that the effects are significantly larger in locations where incumbents faced a much more severe crisis in 2020.

Table 4: Treatment Effect by Covid Incidence in 2020

Covid Incidence in 2020 (Quartile):	1st	2nd	3rd	4th
Baseline (Avg. Under Newcomer)	87.536	92.967	76.180	76.067
Effect (from Reelected Incumbent)	2.639 (5.918)	-3.997 (4.692)	13.980+ (7.208)	23.971* (10.101)
Covid/1,000 (2020, median)	6.17	15.15	24.70	43.16

+p<0.1, *p<0.05. The effect corresponds to β_1 in equation 1. Standard errors are heteroskedasticity-robust. The Baseline is the average outcome for the control group at the cutoff.

CONCLUSION, GENERALIZABILITY, AND POLICY IMPLICATIONS

This article presents novel evidence that politicians' previous time in office significantly affects their performance during a public health crisis. It uses COVID-19 vaccinations in Brazil to show that the policy coverage in locations governed by reelected incumbents was 11% higher than in cities governed by inexperienced newcomers. In addition, the effects cannot be explained by differential experience of bureaucrats or reelection incentives, but are closely related to the level of unique expertise acquired by the mayor in 2020 during the early months of the pandemic.

Crises often have unique features, which can make it difficult to identify the uniting principles (Slough and Tyson, 2025) that generalize the results to other settings. However, our findings likely extend beyond crises to any context marked by increased complexity or sudden changes in procedures required for policy implementation. Our core argument is that political experience generates office-specific knowledge that improves future performance. When policy implementation becomes more complex, the skills required for success become more specialized, while outside knowledge becomes less relevant. In such environments, the benefits of political tenure—and thus accumulated experience—should be more pronounced.

This logic applies not only to crises but also to situations where governments must implement new policies, address uncharted administrative areas, or adapt to institutional changes. One example is the global expansion of cash transfer programs, now a central tool for poverty alleviation in developing nations. In Brazil—the context of our study—this program required rapid, large-scale registration of the targeted population, while also imposing new demands on the public health and education systems, which significantly reshaped mayors’ responsibilities. While existing research examines how incentives related to term-limits affect program implementation (Bueno, 2018; Frey, 2021), the role of political experience remains unexplored.

Under the same logic, the fact that our findings may not always generalize to “normal” times is a key insight of the article. To illustrate this, we replicate in the appendix our RD estimates of the effect of time in office on vaccination using the 2016 electoral cycle. Here, the outcome largely reflects routine immunizations, as the 2017 vaccination lacked both the urgency and the scale of the pandemic-related effort in 2021. Although the estimated coefficient remains positive, it is smaller in magnitude and not statistically significant (Table B.15). This suggests that time in office may be a weak predictor of policy performance in normal conditions, which may help explain why the topic has received limited attention in the public service delivery literature. In this context, our results highlight the importance of studying not only this topic but also other political dynamics in periods of institutional stress. The insights the literature has derived from normal periods may not carry over to periods of heightened pressures on the public sector, which are not uncommon in the developing world, and are precisely when experience is valuable.

These findings also call for future research that searches for factors that can mitigate the negative impact of limited on-the-job experience during crises. We provide suggestive evidence on this point in Appendix A, where we estimate heterogeneous incumbency effects by mayoral education, gender, prior (non-consecutive) time in office, and local bureaucratic capacity in health care.³³ The vaccination gap is smaller among more educated and female mayors, indicating that lack of experience might be more consequential for men and less educated politicians: newcomers

³³ Although the estimated differences in effects across groups are not statistically significant at the usual confidence levels, some are large in magnitude and fairly consistent with the findings of the related literature.

in these groups lag incumbents by 6 p.p. and 11 p.p., respectively. The gender result aligns with evidence that female mayors performed better in early pandemic containment in Brazil ([Bruce et al., 2022](#); [Chauvin and Tricaud, 2024](#)). The education result is also consistent with the literature’s findings that more educated mayors are better at securing federal resources and managing personnel spending ([Desai and Frey, 2023](#); [Rocha, Fernandez Orellano, and Bugarin, 2018](#)).

Also, while newcomers with past (pre-2016) mayoral experience still lag reelected incumbents in vaccinations (a 5.6 p.p. effect), the gap is smaller than the one observed against newcomers that were never mayors before (a 10.4 p.p. effect). This suggests that political experience in the same office, even if it is not recent or acquired during a crisis, might still moderate performance in critical periods. The effects are also slightly higher in municipalities that had a smaller bureaucracy dedicated to health care services, suggesting that the effects of experience (and lack thereof) might be magnified in environments of poor infrastructure.

Future research should further investigate which mechanisms and dimensions of experience drive better performance. Our mediation analysis suggests that the management and selection of bureaucrats may constitute one relevant, but far from exhaustive, channel. Other possibilities include demand-side mechanisms, such as greater trust in more experienced incumbents, which could increase the population’s willingness to vaccinate. Ultimately, our setting does not allow us to precisely disentangle the range of mechanisms that may underlie these effects.

Finally, within a broader literature that has emphasized politicians’ willingness to implement policies, our findings should also motivate studies focusing on how different dimensions of ability shape policy, similar to work that focuses on incumbents’ educational background ([Carnes and Lupu, 2016](#)) or bureaucracy quality ([Slough, 2024a](#)).

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Competing Interests

The author(s) declare none

APPENDIX FOR ONLINE PUBLICATION

When Crisis Hits, Does Experience Matter?

Evidence from COVID-19 vaccination

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A Compensating for Lack of Mayoral Experience

We have shown evidence that the unique experiences of incumbent politicians during a public health crisis allowed them to outperform newcomers in the implementation of a critical public policy. However, rather than interpreting these results as evidence that voters should always choose experienced politicians in times of crises, we believe that these findings motivate future research that attempts to understand how the adverse effects of lack of experience can be compensated for by other traits of candidates or the institutional environment.

In an attempt to do so, we explore other traits of mayors or municipalities that could moderate the effects of the experience documented in this article. We emphasize that these heterogeneity exercises do not have the same internal validity as the main RD results and do not allow us to establish precise causal links between effects and the moderating variables. They are only informative to the extent that their patterns are broadly consistent with the arguments previously made by the related literature.

Table A.1 shows the results. In every column, we split our main sample (Mayor Turnover) into two subsamples, according to the profile of the elected mayors or municipalities based on each characteristic of interest, and estimate the effects for both groups. Column (1) compares municipalities governed by mayors with and without a college education; column (2) shows the differential effects by the mayor's gender; column (3) compares reelected incumbents to a control group composed of newcomers that had previously been mayors in past tenures (i.e., before 2016);¹ and column (4) shows the differential effects for municipalities with the size of the health care bureaucracy below and above the median.

Although we find sizable differences in magnitudes for the effects across groups, none of these differences are statistically significant. Nevertheless, the direction of the estimates is in line with both our argument and the previous findings from the related literature.

¹Brazil limits executive officeholders to two consecutive terms. This means that mayors who served one or more terms before 2016 could have returned to the post in the 2020 election, as long as they were not the elected mayor in 2016.

Table A.1: Effect Heterogeneity by the Profile of Mayors and Bureaucracies

	HETEROGENEITY			
	I	II	III	IV
Group A:	Mayor without a Bachelor's Degree	Male Mayors	Mayors without Previous Experience	WEAK Health Care
Effect	14.393* (4.749)	8.225+ (4.377)	10.391* (4.886)	14.078* (4.648)
Municipalities	171	309	253	180
Group B:	Mayor with a Bachelor's Degree	Female Mayors	Mayors with Previous Experience	STRONG Health Care
Effect	3.825 (5.980)	1.780 (8.185)	5.553 (6.242)	4.478 (5.985)
Municipalities	191	53	109	182
Difference (A-B)	-10.567 (7.636)	-6.445 (9.282)	-4.838 (7.927)	-9.600 (7.578)
Bandwidth	2.74	2.74	2.74	2.74

+p<0.1, *p<0.05. The coefficients show the effect of an incumbent's victory for each subsample of the *Mayor Turnover* sample. All variables are explained in the text. Coefficients in each column are estimated by a single regression where we interact the binary variable in the column title with T_i , D_i , and $T_i \times D_i$ in equation 1.

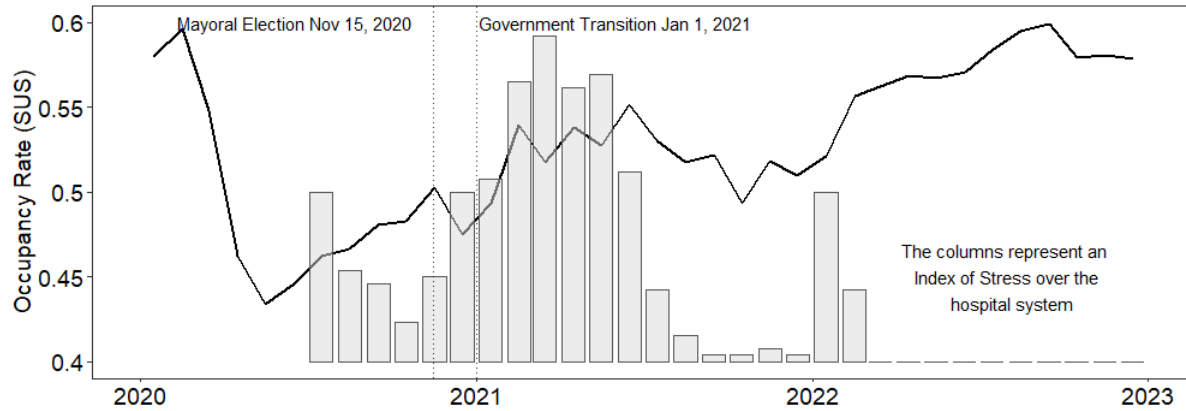
First, the vaccination gap between incumbents and newcomers is low among more educated mayors and among female mayors. In other words, the lack of experience is much more problematic for the performance of men and uneducated politicians, as newcomers in these groups lag incumbents by 6 p.p. and 11 p.p., respectively. The gender result is in line with other studies that show how female mayors performed better in containment measures during the first few months of the pandemic in 2020 in Brazil (Bruce et al., 2022; Chauvin and Tricaud, 2024). For education, the weaker finding is also consistent with previous studies that show mixed results in the country: while education does not affect corruption levels (Carnes and Lupu, 2016), more

educated mayors seem to be better at obtaining federal resources and managing personnel spending ([Desai and Frey, 2023](#); [Rocha, Fernandez Orellano, and Bugarin, 2018](#)).

Also, while newcomers with past (pre-2016) mayoral experience still lag reelected incumbents in vaccinations (a 5.6 p.p. incumbency effect), the gap is smaller than the one observed against newcomers that were never mayors before (a 10.4 p.p. incumbency effect). This suggests that political experience in the same post, even if it is not during a crisis, might moderate performance in more critical periods. The effects are also slightly higher in municipalities that, pre-treatment, had a smaller bureaucracy dedicated to health care services. This suggests that the effects of experience (and lack thereof) might be magnified in environments with poor infrastructure, where the demand for public services vastly exceeds the supply.

B Figures and Tables

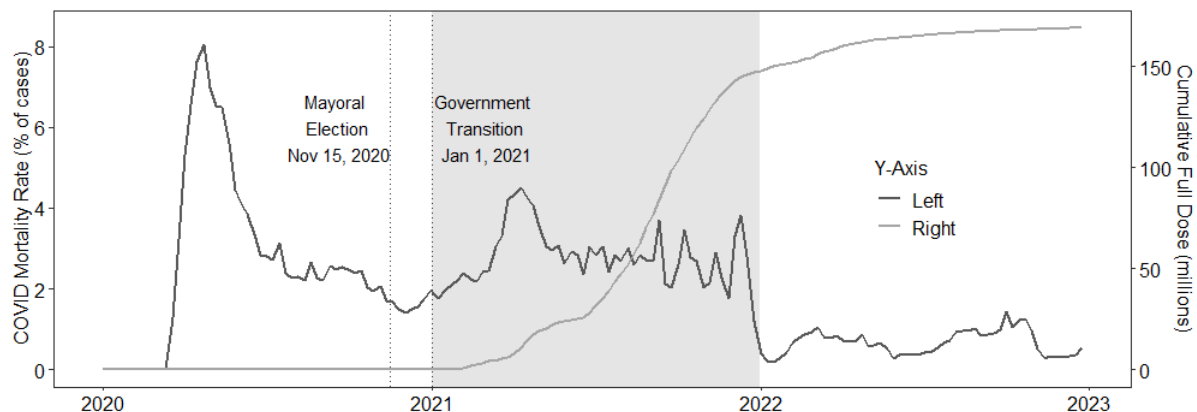
Figure B.1: Hospital Occupancy Rates and System Stress



The line represents the overall occupancy rate in hospital beds in the public system (SUS). The data comes from tab-net.datasus.gov.br.

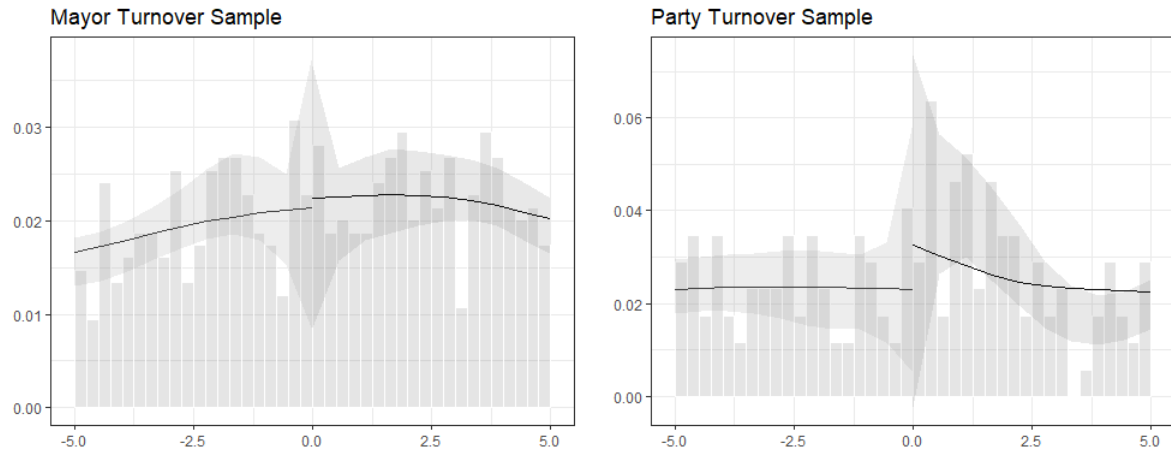
The columns show an index of the stress put onto the occupancy of the entire system, calculated by FioCruz (fiocruz.br). The series starts in July 2020. The index is calculated using the estimated occupancy in ICU beds across the 26 Brazilian states, at the end of each month. FioCruz classified states as highly critical when occupancy was above 80%; as moderately critical when occupancy was above 60%; and not critical when it was lower than 60%. The index is the number of highly critical minus the number of not critical states. For the plot, its value has been normalized to fit the scale, so it has no substantive interpretation. For perspective, the highest column (March 2021) had an actual value of 24, with 24 states in critical condition, 2 rated as moderate, and none as not critical. The lowest possible index value is -26 (at the end of 2022).

Figure B.2: COVID Mortality Rate and Vaccines in Brazil



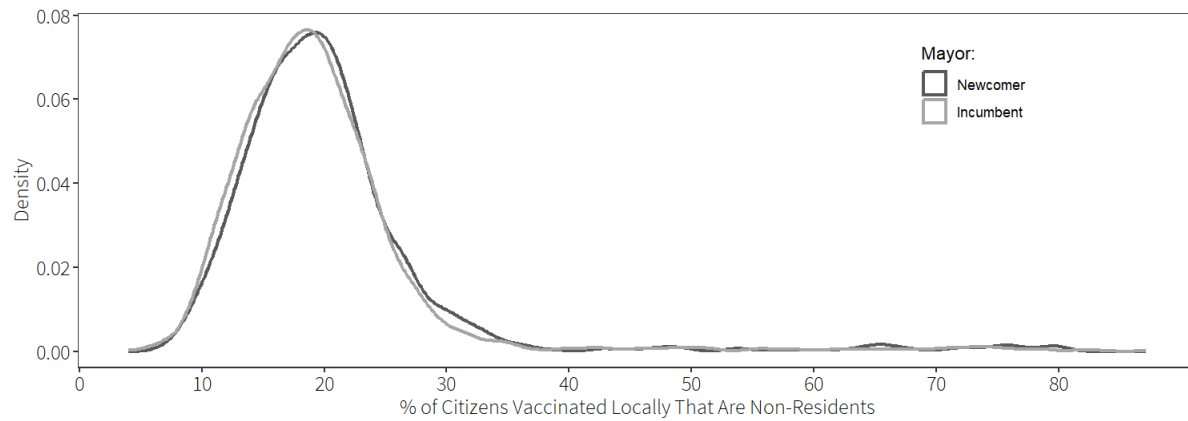
The shaded area represents the year 2021, which is the focus of this analysis. Data on COVID-19 infections and vaccines come from [MS \(2020\)](#) and [MS \(2024\)](#), respectively.

Figure B.3: RD Density Test for both Samples



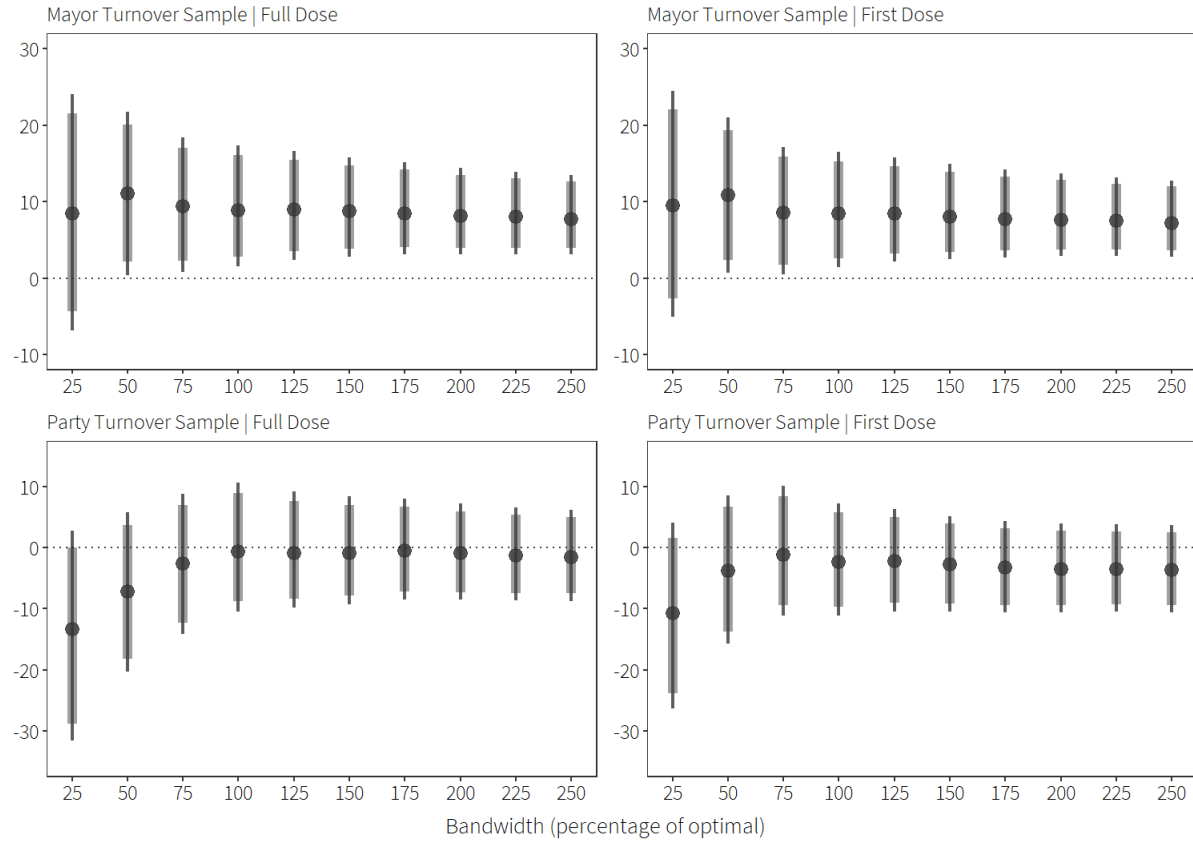
The p-values for the density tests are 0.97 and 0.89, respectively.

Figure B.4: Density of the Share of Eligible Residents that were Vaccinated Locally



The variable of interest accounts for all persons vaccinated in the municipality while also declaring residence in that location. The plot shows the density for each group—municipalities governed by newcomers or reelected incumbents—in our Mayor Turnover Sample.

Figure B.5: Robustness of Effects to Bandwidth Choice



The coefficients correspond to β_1 in equation 1. The thin (wide) bars correspond to the 95% (90%) confidence intervals.

Table B.1: Continuity of Municipal and Candidate Covariates | Party Turnover Sample

	Coef.	S.E.	90% C.I.	Baseline	Band.	Munis.
Registered Voters ('000)	0.425	(0.365)	[-0.16,1.04]	7.143	5.14	87
Municipal GDP	0.059	(0.226)	[-0.34,0.41]	16.246	4.50	72
COVID cases	-0.195	(0.303)	[-0.72,0.28]	19.975	5.26	89
Health Bureaucracy	-2.808	(2.272)	[-6.93,0.55]	12.441	3.51	58
Campaign Spending Cap	0.142	(0.194)	[-0.18,0.46]	0.155	4.13	66
Population Density	0.022	(0.020)	[-0.01,0.06]	0.031	4.72	78
Municipal Deaths in 2020	-0.003	(0.102)	[-0.17,0.17]	8.845	4.54	72
2018 Bolsonaro votes	0.054	(0.075)	[-0.07,0.17]	0.380	5.06	83
Mayor Race	0.096	(0.169)	[-0.17,0.38]	0.550	4.84	80
Mayor College	0.002	(0.161)	[-0.27,0.26]	0.578	5.69	94
Mayor Left-Wing	0.062	(0.127)	[-0.13,0.29]	0.092	4.45	71
Mayor Health Care	-0.044	(0.090)	[-0.21,0.09]	0.071	3.58	58
Mayor Female	0.179	(0.108)	[0.00,0.35]	0.047	4.81	80
Share of Health Bureaucracy in:						
Primary Care	0.090	(0.069)	[-0.01,0.21]	0.538	3.55	58
Doctors	0.022+	(0.013)	[0.00,0.05]	0.057	4.17	66
Nurses	0.014	(0.034)	[-0.04,0.07]	0.229	5.00	83
Health Agents	0.056	(0.040)	[0.00,0.13]	0.251	3.13	53
Other in Health	-0.072	(0.054)	[-0.17,0.01]	0.417	3.99	63
Managers	-0.012	(0.017)	[-0.04,0.01]	0.040	4.50	72

+p<0.1, *p<0.05. Standard errors (parentheses) and 90% confidence intervals are bias-robust. The *Coefficient* is the difference in the outcome between winners and losers, and the baseline is the average outcome for losers, both at the discontinuity. Bandwidths are optimal.

Variables: (1) Registered voters in 2020 (logged); (2) GDP per capita, average of 2018-2020 in R\$ (logged); (3) COVID cases per 1,000 in 2020 (logged); (4) Municipal bureaucrats (x1,000) working on health services in Oct 2020, per capita; (5) The municipal cap for campaign spending by mayor in 2020, in R\$ million (logged); (6) density ('000 voters divided by km2 area); (7) total deaths in 2020 in the municipality per 1,000 voters (logged); (8) vote share for Bolsonaro in the second round of the 2018 presidential election; (9-13) Binary variables that indicate whether the mayoral candidate is white (9); has a 4-year, university degree (10); belongs to a Left-wing party (11) – Left-wing parties are PDT, PSB, PCdoB, PSOL, PT, and CIDADANIA; is a health care worker (12); and is female (13); (14-19) Share of the municipal bureaucracy working in different sub-areas health care, by type: (14) in primary care (doctors, nurses, or health agents); (15) doctors; (16) nurses; (17) health agents; (18) managers; and (19) others.

Table B.2: Continuity of Municipal and Candidate Covariates | Mayor Turnover Sample

	Coef.	S.E.	90% C.I.	Baseline	Band.	Munis.
Registered Voters ('000)	0.113	(0.214)	[-0.22,0.48]	9.589	3.52	220
Municipal GDP	0.140	(0.135)	[-0.08,0.36]	17.691	4.42	271
COVID cases	0.260	(0.251)	[-0.12,0.70]	13.526	2.77	175
Health Bureaucracy	0.554	(0.972)	[-0.90,2.29]	11.227	2.64	169
Campaign Spending Cap	0.125	(0.121)	[-0.06,0.34]	0.159	3.02	194
Population Density	0.069	(0.053)	[-0.02,0.16]	0.022	3.01	194
Municipal Deaths in 2020	0.070	(0.064)	[-0.03,0.18]	8.545	3.23	206
2018 Bolsonaro votes	0.037	(0.041)	[-0.03,0.10]	0.440	4.71	279
Mayor Race	0.086	(0.101)	[-0.08,0.25]	0.635	3.63	226
Mayor College	-0.081	(0.112)	[-0.27,0.10]	0.478	3.32	210
Mayor Left-Wing	0.067	(0.091)	[-0.07,0.23]	0.154	2.93	191
Mayor Health Care	0.043	(0.048)	[-0.03,0.13]	0.032	4.05	246
Mayor Female	-0.154+	(0.093)	[-0.31,-0.01]	0.295	2.98	193
Share of Health Bureaucracy in:						
Primary Care	-0.021	(0.025)	[-0.06,0.02]	0.631	2.71	173
Doctors	0.014	(0.011)	[0.00,0.03]	0.070	2.85	179
Nurses	0.006	(0.016)	[-0.02,0.03]	0.250	2.61	168
Health Agents	-0.038	(0.027)	[-0.09,0.00]	0.309	2.35	157
Other in Health	0.016	(0.024)	[-0.02,0.06]	0.345	2.57	166
Managers	0.003	(0.006)	[-0.01,0.01]	0.025	3.44	218

+p<0.1, *p<0.05. Standard errors (parentheses) and 90% confidence intervals are bias-robust. The *Coefficient* is the difference in the outcome between winners and losers, and the baseline is the average outcome for losers, both at the discontinuity. Bandwidths are optimal.

Variables: (1) Registered voters in 2020 (logged); (2) GDP per capita, average of 2018-2020 in R\$ (logged); (3) COVID cases per 1,000 in 2020 (logged); (4) Municipal bureaucrats (x1,000) working on health services in Oct 2020, per capita; (5) The municipal cap for campaign spending by mayor in 2020, in R\$ million (logged); (6) density ('000 voters divided by km2 area); (7) total deaths in 2020 in the municipality per 1,000 voters (logged); (8) vote share for Bolsonaro in the second round of the 2018 presidential election; (9-13) Binary variables that indicate whether the mayoral candidate is white (9); has a 4-year, university degree (10); belongs to a Left-wing party (11) – Left-wing parties are PDT, PSB, PCdoB, PSOL, PT, and CIDADANIA; is a health care worker (12); and is female (13); (14-19) Share of the municipal bureaucracy working in different sub-areas health care, by type: (14) in primary care (doctors, nurses, or health agents); (15) doctors; (16) nurses; (17) health agents; (18) managers; and (19) others.

Table B.3: Effects with Covariates or a Quadratic Polynomial (Party Turnover Sample)

	Full Dose	1st Dose	Full 2021+2022
Specification: Includes Covariates			
Treatment Effect	6.877* (2.954)	5.851* (3.060)	7.143* (3.330)
C.I. (95%)	[1.56,13.14]	[0.19,12.19]	[1.08,14.13]
Bandwidth	2.74	2.80	2.84
Municipalities	362	367	371
Specification: Includes Covariates and Fixed Effects for Party and State			
Treatment Effect	4.515* (2.371)	4.749* (2.587)	4.400+ (2.686)
C.I. (95%)	[0.26,9.55]	[0.05,10.19]	[-0.50,10.03]
Bandwidth	2.84	2.92	2.90
Municipalities	371	387	385
Specification: Quadratic Polynomial (No Covariates)			
Treatment Effect	9.321* (4.034)	8.551* (3.879)	9.529* (4.195)
C.I. (95%)	[1.63,17.44]	[1.08,16.29]	[1.46,17.90]
Bandwidth	5.75	5.75	5.60
Municipalities	706	707	692

+p<0.1, *p<0.05. The effect always corresponds to β_1 in equation 1, for each specification, for the Mayor Turnover Sample. Standard errors and 95% confidence intervals are bias-robust. All variables are explained in the text.

Table B.4: Effects by Residence Status

	All (1)	Residents (2)	Non-Residents (3)	Ratio (4)
Treatment Effect	8.883* (4.027)	7.679* (3.304)	1.620 (2.081)	0.707 (2.447)
C.I. (95%)	[1.54,17.33]	[1.57,14.52]	[-2.39,5.77]	[-3.94,5.65]
Baseline	83.704	67.408	16.413	79.657
Bandwidth	2.74	4.17	3.31	3.09
Municipalities	362	543	430	410

+p<0.1, *p<0.05. The effect always corresponds to β_1 in equation 1, for each specification, for the Mayor Turnover Sample. Standard errors and 95% confidence intervals are bias-robust.

Outcome definition: Column (1) uses the total number of full vaccine doses given in 2021 in the municipality (our main specification); column (2) uses the full doses given in 2021 to residents of the municipality; column (3) uses the full doses given in 2021 to non-residents of the municipality. In all three cases, the number of vaccines is shown as a percentage of the total number of registered voters. Column (4) has the percentage of total doses, in 2021, that were given to residents.

Table B.5: Effects with Different Proxies for the Vaccine Target Population

	Registered Voters (2020)	Actual Voters (2020)	Total Population (2010)
Treatment Effect	8.883*	12.729*	7.774*
	(4.027)	(5.486)	(3.276)
C.I. (95%)	[1.54,17.33]	[2.68,24.18]	[1.84,14.68]
Baseline	83.704	101.216	68.600
Bandwidth	2.74	2.74	2.69
Municipalities	362	362	352

+p<0.1, *p<0.05. The effect always corresponds to β_1 in equation 1, for each specification, for the Mayor Turnover sample. Standard errors and 95% confidence intervals are bias-robust. All specifications use the total number of full vaccine doses given in 2021 in the municipality.

Target Population: Column (1) uses the number of registered voters who turned out in the 2020 elections (our main specification); column (2) uses the number of actual voters in 2020; and column (3) uses the municipal population from the most recent census (2010).

Table B.6: Effects on Deaths

Outcome	DEATHS IN 2021		DEATHS IN 2021–2022	
Covariates?	No	Yes	No	Yes
Mayor Turnover Sample				
Treatment Effect	-0.102+	-0.116*	-0.082+	-0.092*
	(0.061)	(0.060)	(0.047)	(0.047)
C.I. (95%)	[-0.22,0.02]	[-0.23,0.00]	[-0.18,0.01]	[-0.19,0.00]
Baseline	1.369	1.370	1.275	1.275
Bandwidth	3.60	3.58	3.57	3.58
Municipalities	467	464	464	465
Party Turnover Sample				
Treatment Effect	0.029	-0.001	0.037	0.021
	(0.127)	(0.120)	(0.097)	(0.089)
C.I. (95%)	[-0.23,0.26]	[-0.24,0.23]	[-0.16,0.22]	[-0.16,0.19]
Baseline	1.280	1.280	1.203	1.203
Bandwidth	3.91	3.83	4.08	4.08
Municipalities	141	140	145	145

+p<0.1, *p<0.05. The outcome is also the annual deaths in the municipality (period average for 2021–2022) divided by the total deaths in 2019 (pre-COVID). The effect always corresponds to β_1 in equation 1, for each specification. Standard errors and 95% confidence intervals are bias-robust. The included covariates are listed in Table B.1.

Table B.7: Effect including large cities with two-round elections

	Full Dose	1st Dose	Full in 2021+2022
Treatment Effect	8.934*	8.502*	9.180*
	(4.014)	(3.849)	(4.102)
C.I. (95%)	[1.62,17.35]	[1.44,16.53]	[1.68,17.75]
Baseline	83.662	96.465	96.671
Effect over Baseline	10.68%	8.81%	9.50%
Bandwidth	2.76	2.78	2.82
Municipalities	365	367	372

+p<0.1, *p<0.05. The effect corresponds to β_1 in equation 1, for the **MAYOR Turnover Sample**. Standard errors and 95% confidence intervals are bias-robust. The Baseline is the average outcome for the municipalities that elected newcomers (control group) at the cutoff.

Table B.8: Effect Heterogeneity by Ideology of Mayor and Voters

Mayor	LEFT-WING			RIGHT-WING		
Median Voter	Left	Moderate	Right	Left	Moderate	Right
Incumbency Effect	31.358*	26.995*	4.296	4.378	8.944	5.443
	(10.360)	(13.439)	(9.906)	(5.547)	(7.680)	(5.919)
Effective Municipalities	29	21	18	96	104	107
Bolsonaro Vote (2018)	19.6	47.0	69.2	19.5	46.3	69.8

+p<0.1, *p<0.05. Standard errors (parentheses) and 90% confidence intervals are bias-robust. The *Coefficient* is the difference in the outcome between winners and losers. Bandwidth is optimal. Here we only use municipalities that elected a mayor from one of the 16 large parties for which we have ideological scores (91% of the sample). The last row has the average share of the municipal vote for Bolsonaro in the 2018 election. Effective municipalities are the ones with close elections, as defined by the optimal bandwidth. All coefficients come from a single regression where indicators for the mayor and voter ideology are interacted with the treatment dummy and themselves at the discontinuity.

Table B.9: RD Effects: Dismissals and New Hires

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Dismissals							
Party Turnover	-3.224*	-1.328*	-0.014	-0.823*	-0.492*	-0.202*	-1.694*
	(1.183)	(0.485)	(0.101)	(0.275)	(0.219)	(0.075)	(0.815)
Mayor Turnover	-1.426*	-0.539*	0.053	-0.369*	-0.223*	-0.136*	-0.751*
	(0.376)	(0.212)	(0.090)	(0.115)	(0.103)	(0.047)	(0.224)
Difference (M - P)	1.798	0.790	0.067	0.454	0.269	0.066	0.942
	(1.241)	(0.530)	(0.136)	(0.298)	(0.242)	(0.088)	(0.846)
New Hires							
Party Turnover	-2.519*	-0.834	0.110	-0.563+	-0.381+	-0.114	-1.571*
	(0.874)	(0.511)	(0.112)	(0.305)	(0.227)	(0.098)	(0.438)
Mayor Turnover	-0.975+	-0.241	0.172	-0.275*	-0.137	-0.059	-0.675*
	(0.505)	(0.246)	(0.120)	(0.135)	(0.108)	(0.044)	(0.344)
Difference (M - P)	1.544	0.593	0.062	0.288	0.244	0.055	0.896
	(1.009)	(0.568)	(0.164)	(0.334)	(0.252)	(0.107)	(0.557)
Municipalities (P)	131	131	131	131	131	131	131
Municipalities (M)	431	431	431	431	431	431	431

+p<0.1, *p<0.05. The coefficients correspond to β_1 in equation 1, for each sample and specification as indicated. Standard errors and 95% confidence intervals are heteroskedasticity-robust. The outcomes are always defined as the number of health care workers in each category per 1,000 voters. Hires and dismissals correspond to the period between Oct 2020 and Dec 2021. **Outcomes by column.** (1): total health care workers; (2): total primary care workers (doctors + nurses + health agents); (3): doctors; (4): nurses; (5): health agents; (6): managers; (7): other health care workers.

Table B.10: RD Effects: Turnover and Net Hires

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Turnover							
Party Turnover	-5.743*	-2.162*	0.097	-1.386*	-0.873*	-0.316*	-3.264*
	(1.905)	(0.950)	(0.179)	(0.559)	(0.417)	(0.151)	(1.138)
Mayor Turnover	-2.401*	-0.779+	0.225	-0.644*	-0.360+	-0.195*	-1.427*
	(0.822)	(0.422)	(0.188)	(0.230)	(0.197)	(0.084)	(0.505)
Difference (M - P)	3.342	1.383	0.128	0.742	0.512	0.121	1.838
	(2.075)	(1.040)	(0.260)	(0.604)	(0.461)	(0.173)	(1.245)
Net Hires							
Party Turnover	0.706	0.495	0.124	0.260	0.111	0.088	0.123
	(0.834)	(0.302)	(0.116)	(0.159)	(0.160)	(0.087)	(0.646)
Mayor Turnover	0.452	0.298	0.119	0.093	0.086	0.077*	0.076
	(0.343)	(0.183)	(0.098)	(0.099)	(0.074)	(0.036)	(0.286)
Difference (M - P)	-0.254	-0.197	-0.005	-0.167	-0.025	-0.011	-0.047
	(0.902)	(0.353)	(0.152)	(0.187)	(0.176)	(0.094)	(0.707)
Municipalities (P)	131	131	131	131	131	131	131
Municipalities (M)	431	431	431	431	431	431	431

+p<0.1, *p<0.05. The coefficients correspond to β_1 in equation 1, for each sample and specification as indicated. Standard errors and 95% confidence intervals are heteroskedasticity-robust. The outcomes are always defined as the number of health care workers in each category per 1,000 voters. Turnover is the sum of hires and dismissals, and net hires is the difference between them. Hires and dismissals correspond to the period between Oct 2020 and Dec 2021. **Outcomes by column.** (1): total health care workers; (2): total primary care workers (doctors + nurses + health agents); (3): doctors; (4): nurses; (5): health agents; (6): managers; (7): other health care workers.

Table B.11: RD Effects on Vaccine Acquisition

Outcome	DOSES RECEIVED UNTIL DECEMBER 2022		FULL SCHEDULE VACC. PER DOSE RECEIVED	
	No	Yes	No	Yes
Covariates?				
Treatment Effect	-3.168	-6.842	27.333*	30.758*
	(8.893)	(7.940)	(12.894)	(13.125)
C.I. (95%)	[-21.00,13.86]	[-23.04,8.09]	[8.46,59.01]	[11.50,62.95]
Baseline	213.105	213.105	33.583	33.583
Bandwidth	3.36	3.36	2.71	2.71
Municipalities	427	427	347	347

+p<0.1, *p<0.05. The first outcome considers the total number of doses received between January 2021 and December 2022, as a % share of registered voters. The second outcome is the total number of people covered by the full vaccination schedule in 2021–2022, as a % of doses received. The effect always corresponds to β_1 in equation 1, for each specification, and . Standard errors and 95% confidence intervals are bias-robust.

Table B.12: Profiles of the Close Election Samples

SAMPLE: Close Elections?	MAYOR TURN.		PARTY TURN.	
	No <i>Mean</i>	Yes <i>Difference</i>	No <i>Mean</i>	Yes <i>Difference</i>
Registered Voters ('000)	16.297	1.303	15.193	-0.536
Municipal GDP	22.764	2.901*	25.680	-0.212
2020 COVID cases	24.322	0.600	24.245	-1.021
Health Bureaucracy	11.103	0.135	10.950	-0.345
Campaign Spending Cap	0.197	0.026*	0.183	0.002
Population Density	0.044	0.013	0.040	0.002
Deaths in 2020	9.084	0.188	9.182	0.100
2018 Bolsonaro votes	0.446	0.008	0.445	-0.010
Mayor Race	0.674	-0.016	0.637	0.044
Mayor College	0.528	0.010	0.506	0.075+
Mayor Left-Wing	0.155	0.020	0.206	-0.029
Mayor Health Care	0.083	-0.007	0.056	0.015
Mayor Female	0.146	-0.036+	0.119	0.019
Share of Health Bureaucracy in:				
Primary Care	0.626	-0.003	0.612	0.009
Doctors	0.074	0.005*	0.073	0.003
Nurses	0.251	0.000	0.239	0.007
Health Agents	0.301	-0.009	0.300	-0.001
Other in Health	0.348	0.004	0.360	-0.008
Managers	0.026	0.000	0.028	-0.001
Municipalities	362	2625	160	530

+p<0.1, *p<0.05. The columns labeled *Mean* show the average value of each variable for the group of municipalities that did not have a close election in 2020. The columns labeled *Difference* show the average difference between this and the close election sample. The detailed description of each variable can be found in Table B.1.

Table B.13: Profiles of the Party and Mayor Turnover Samples

SAMPLE:	ALL MUNICIPALITIES		CLOSE ELECTIONS	
	Party Turn.	Mayor Turn.	Party Turn.	Mayor Turn.
	<i>Mean</i>	<i>Difference</i>	<i>Mean</i>	<i>Difference</i>
Number of Voters (per '000)	14.781	2.662*	16.081	0.120
Municipal GDP	25.516	-0.203	24.285	-1.377
2020 COVID cases	23.461	1.388+	24.554	-0.502
Health Bureaucracy	10.685	0.537*	11.060	0.173
Campaign Expenditure Cap	0.185	0.035*	0.187	0.009
Population Density	0.042	0.013*	0.041	0.004
Municipal Deaths in 2020	9.259	-0.010	9.124	-0.020
2018 Bolsonaro votes	0.438	0.016	0.434	0.010
Mayor Race	0.671	-0.011	0.634	0.030
Mayor College	0.564	-0.028	0.525	-0.001
Mayor Left-Wing	0.184	-0.012	0.177	-0.010
Mayor Health Care	0.068	0.008	0.053	0.014
Mayor Female	0.133	-0.019	0.125	0.025
Share of Health Bureaucracy in:				
Primary Care	0.619	0.004	0.605	0.016
Doctors	0.075	0.004+	0.071	0.004
Nurses	0.244	0.007*	0.238	0.012
Health Agents	0.299	-0.006	0.296	0.001
Other in Health	0.354	-0.003	0.365	-0.013
Managers	0.027	-0.001	0.029	-0.003
Municipalities	690	2987	177	627

+p<0.1, *p<0.05. The columns labeled *Mean* show the average value of each variable for the Party Turnover sample. The columns labeled *Difference* show the average difference between the samples. The detailed description of each variable can be found in Table B.1 (appendix).

Table B.14: Bureaucratic Turnover as a Mediator of Incumbency Effects

Full Regression		First Stage	ADE (β_3)	ACME ($\beta_2\gamma$)
β_3	γ	β_2	% of total effect	
7.496+	-5.832*	-0.219+	7.496	1.277
(3.911)	(2.058)	(0.113)	85%	15%

+p<0.1, *p<0.05.

We follow the convention of [Imai et al. \(2011\)](#) for the equation system and for labeling the coefficients.

The full regression has our main vaccination variable as the outcome, regressed on both the incumbency treatment and the value of the mediator, which is measured as a binary variable indicating whether the total turnover in 2021 (hires + dismissals) in the municipality was above the median:

$$Vacc_i = \alpha_3 + \beta_3 Inc_i + \gamma M_i + \theta_{3a} D_i + \theta_{3b} Inc_i D_i + \theta_{3c} M_i D_i + \epsilon_{3i}.$$

The coefficient β_3 represents the average direct effect (ADE).

The first stage has the mediator regressed on the incumbency treatment, as follows:

$$M_i = \alpha_2 + \beta_2 Inc_i + \theta_{2a} D_i + \theta_{2b} Inc_i D_i + \epsilon_{2i}.$$

The average causal mediation effect (ACME) comes from the multiplication of β_2 and γ .

All regressions are estimated for the Mayor Turnover sample.

Table B.15: RD Effect for Infant Vaccinations in 2017

Sample:	MAYOR TURNOVER		PARTY TURNOVER	
	Total Doses (log)	Doses per capita	Total Doses (log)	Doses per capita
Effect	0.105 (0.206)	5.050 (4.352)	-0.119 (0.394)	-0.009 (5.469)
C.I. (95%)	[-0.28,0.52]	[-3.50,13.56]	[-0.89,0.65]	[-11.41,10.03]
Baseline	8.654	56.853	8.901	58.935
Bandwidth	4.69	4.39	4.88	4.89
Municipalities	560	527	201	202

+p<0.1, *p<0.05. The effect corresponds to β_1 in equation 1. Standard errors and 95% confidence intervals are bias-robust. All variables are explained in the main text. The Baseline is the average outcome for the municipalities that elected newcomers (control group) at the cutoff. Includes all vaccines applied by the public health sector. Doses per capita use the 2010 Census population in the denominator.

C A Discussion on Compensating Differentials

In this section, we assess the potential impact of compensating differentials on our regression discontinuity results, with a focus on the role of unobserved politician ability (or quality), given its relevance to this specific application.

The concern here is that if the pool of incumbents who end up in close elections has, on average, a higher ability than their challengers, then our estimated vaccination effects could reflect this quality advantage (of reelected incumbents) rather than the effect of prior experience. However, this problem would arise only under an important condition: there must exist a candidate-level incumbency disadvantage that depresses incumbents' vote shares sufficiently to push high-quality candidates—who would otherwise win by large margins—into close elections. As a result, candidates in close races would differ not only in incumbency status but also in this compensating differential, which could reflect ability or another unobserved factor that positively affects both vote shares and policy outcomes.

We emphasize that the compensating differential problem arises only if voters observe some candidate characteristic that the researcher cannot measure and reward or penalize politicians for it. Because this compensating variable is unobserved by the researcher, its direct impact on vote shares cannot be directly quantified. Instead, we present three pieces of evidence that mitigate concerns that our results are driven by an ability-based explanation.

First, we examine the incumbent's decision to run for reelection. Approximately 25% of the mayors elected in 2016 who were not term-limited chose not to run again in 2020. A potential concern is that if lower-ability incumbents disproportionately opt out of reelection, the 75% who do run may, on average, have higher ability than the pool of challengers.

Although we do not observe candidate ability, the concern arises only if voters do observe it and reward politicians accordingly. If this were the case, the 2016 vote shares should partly capture this dimension of the incumbent's profile. We can therefore assess the role of ability in shaping the pool of candidates by examining whether the incumbent's decision to run again in 2020 is correlated with their 2016 margin of victory. Overall, we find no significant correlation.

The average 2016 margin of victory among incumbents who did not run again was 17.53%, which is very close to the 17.76% margin observed among the incumbents who did run. A t-test for the difference in means yields a p-value of 0.76.

Second, we use the margin of victory in the previous election (2016) to further assess the estimated treatment effects. Following the argument above, mayors elected in 2016 in close races were less likely to have been elected because of superior ability. We emphasize that every incumbent running in 2020 had first been elected as a newcomer in 2016, since Brazil imposes a term limit of two consecutive terms on mayors. If an ability-based explanation were driving our results, then the RD effects estimated for the subset of municipalities that also had close elections in 2016 should be weaker, given the likely profile of this group of incumbents.

Table C.1 shows that this is not the case. We estimate heterogeneous effects for subsamples of municipalities based on the level of competition in the previous mayoral race—specifically, whether the margin of victory was above or below 5%. Although the estimates are not statistically different from one another, the effects for municipalities with close races in 2016 are in fact slightly larger in magnitude, reaching 16.5% of the baseline (column 2). This pattern further suggests that our vaccination results are unlikely to be driven by differences in candidate ability.

Table C.1: Heterogeneity Based on the Margin of Victory in 2016

	Full Sample	Margin of Victory in 2016	
		Below 5%	Above 5%
Treatment Effect	8.883*	12.921*	7.740+
	(4.027)	(6.461)	(4.245)
C.I. (95%)	[1.54,17.33]	[1.09,26.41]	[0.00,16.64]
Baseline	83.704	78.355	85.554
Effect over Baseline	10.61%	16.49%	9.05%
Bandwidth	2.74	4.03	3.58
Municipalities	362	141	339

+p<0.1, *p<0.05. The effect always corresponds to β_1 in equation 1, for each specification, for the Mayor Turnover sample. Standard errors and 95% confidence intervals are bias-robust. All specifications use the total number of full vaccine doses given in 2021 in the municipality.

Finally, we assess the plausibility of a candidate-level incumbency disadvantage in our context, which is a necessary condition for an ability-based explanation of our results. Although previous research has documented an incumbency disadvantage in Brazilian mayoral elections (Klašnja and Titiunik, 2017), those estimates are at the party level and are largely driven by lame-duck mayors who are term-limited. Neither that mechanism nor those findings directly apply to the reelection-eligible incumbents we study here. This raises the question of how plausible it is that, in our sample, the true candidate-level incumbency “advantage” is negative (i.e., a disadvantage).

To address this issue, we conduct a sensitivity analysis to assess how strong the relationship between unobserved ability and vote shares would need to be in our context to generate the concern described above. Specifically, we use the method proposed by Oster (2019), which has been widely used in the social sciences to evaluate the potential role of unobserved variables in the stability of treatment effects. We begin by estimating Equation C.1 below.

$$VoteShare2020_{ic} = \alpha + \beta Inc_{ic} + \theta X_{ic} + \epsilon_{ic} \quad (C.1)$$

where for candidate c in location i , we estimate the relationship between incumbency status (Inc_{ic}) and vote shares ($VoteShare2020_{ic}$) in the 2020 mayoral race. To establish an empirical correspondence with the measure of margin of victory used in RD designs, we normalize the vote shares by the total votes obtained only by these top two candidates. Thus, β corresponds to the average margin of victory of incumbents in these elections. Finally, X_{ic} are pre-determined covariates that include the age, education, gender, and race of the candidate; state and party fixed-effects; and the local margin of victory in the previous race (2016).

The results are shown in Table C.2. The first column shows the baseline estimate of β without any covariates: on average, incumbents sustain a 10% margin of victory over newcomers, which is also statistically significant. Column 2 shows that, after including covariates, the β estimate remains extremely stable at 9.7%, and the R-squared on the regression increases to 0.14 from 0.11 in the baseline.

Table C.2: Sensitivity of the Incumbent Effect on Votes Shares

	Baseline effect (1)	Controlled effect (2)	Bias-adjusted β $R_{max} = .170$ $R_{max} = 0.182$ (3) (4)		Max R_{max} for $\beta = 0$ (5)
Baseline	0.450				
Effect	0.100* (0.003)	0.097* (0.004)	0.094	0.093	1.127
R-Squared	[0.110]	[0.140]			

+p<0.1, *p<0.05. The effect is the estimate of β in equation C.1, for each specification. Standard errors are heteroskedasticity-robust. The regression for controlled effects includes the following covariates: the candidate's age, gender, education (indicator for 4-year bachelor's degree), and race (indicator for white). It also controls for the municipal margin of victory in the previous mayoral race (2016), and it includes fixed effects for the party of the candidate and for states. The bias-adjusted β is calculated under the assumption that the maximum R-squared of the full regression would be $R_{max} = R_{con} + (R_{con} - R_{bas})$, for column 3, and $R_{max} = R_{con} * 1.3$, for column 4, in line with the recommendation by Oster (2019). The last column shows the value of the R-squared in the full regression that would make $\beta = 0$. In all regressions, the implicit λ parameter from Oster (2019) is assumed to be 1, as recommended.

Although the strong and positive β is consistent with a candidate-level incumbency advantage, this coefficient alone does not establish that conclusion. The estimate may bundle the potential incumbency advantage—or even a disadvantage—with other factors correlated with both incumbency status and vote shares, including possible compensating differentials such as politicians' ability. Therefore, this sensitivity analysis considers a scenario in which the compensating differential could be measured, so that β would reflect only the incumbency disadvantage. In other words, we assess how strong the effect of unobserved ability on vote shares would have to be for β to equal zero in this context.

Columns (3) and (4) present the estimated values of β after adjusting for the potential bias introduced by the unobserved compensating differential, based on assumptions about how much of the total variation in the outcome could be explained by the unobserved variable (i.e., through changes in the regression's R^2). Following the recommendations in Oster (2019), we set plausible values for the maximum R^2 and show that, under these conditions, the adjusted β —which under these assumptions reflects only the incumbency advantage—remains positive and highly stable,

at above 9%.

Perhaps the most important result appears in column (5), which reports the value of the R^2 in the hypothetical full regression (including the unobserved ability measure) required to drive the coefficient β to zero. The estimate is 1.13. This implies that any unobserved compensating differential would need to explain more than 100% of the remaining variation in vote shares, which is highly implausible in our context.

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